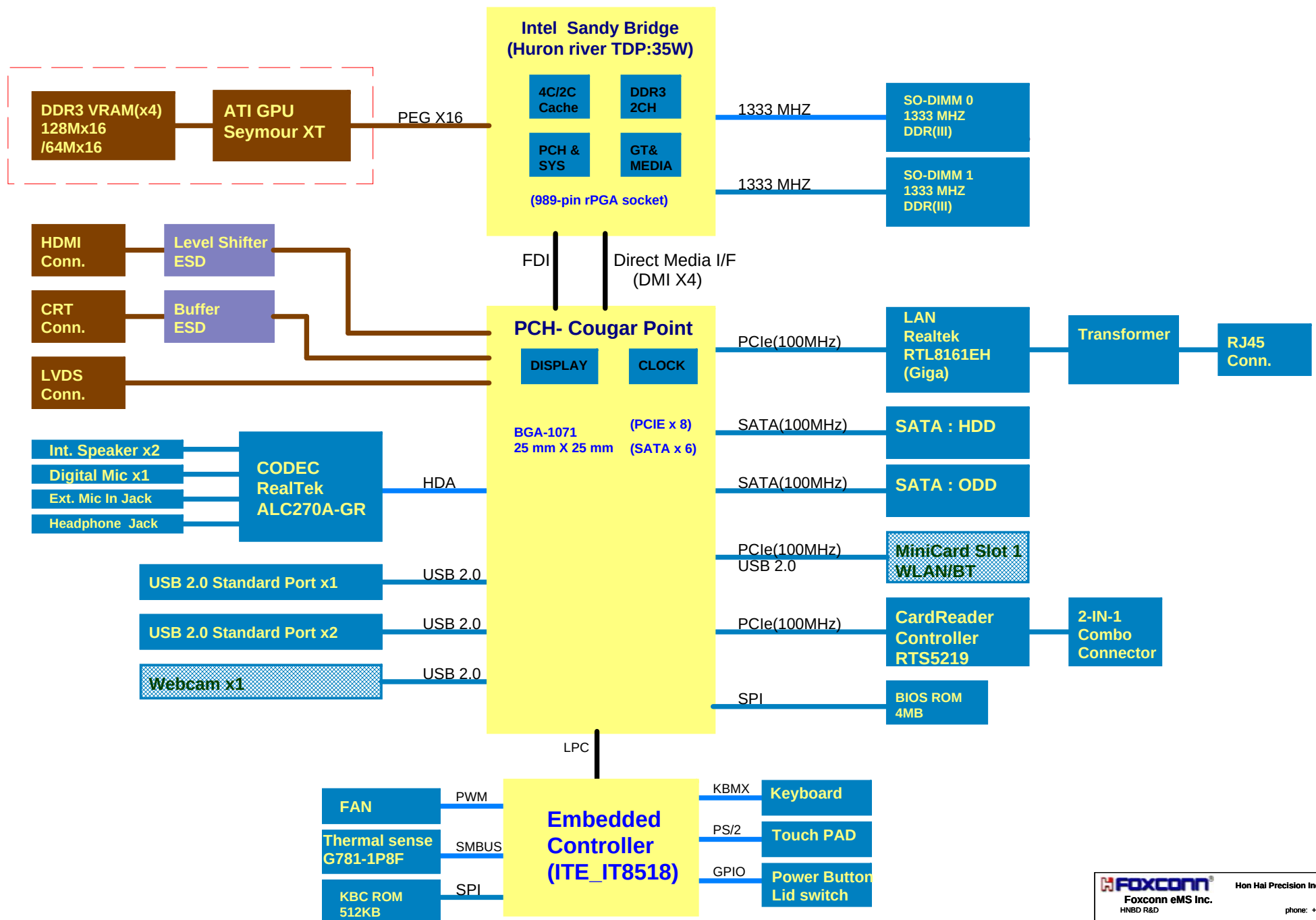


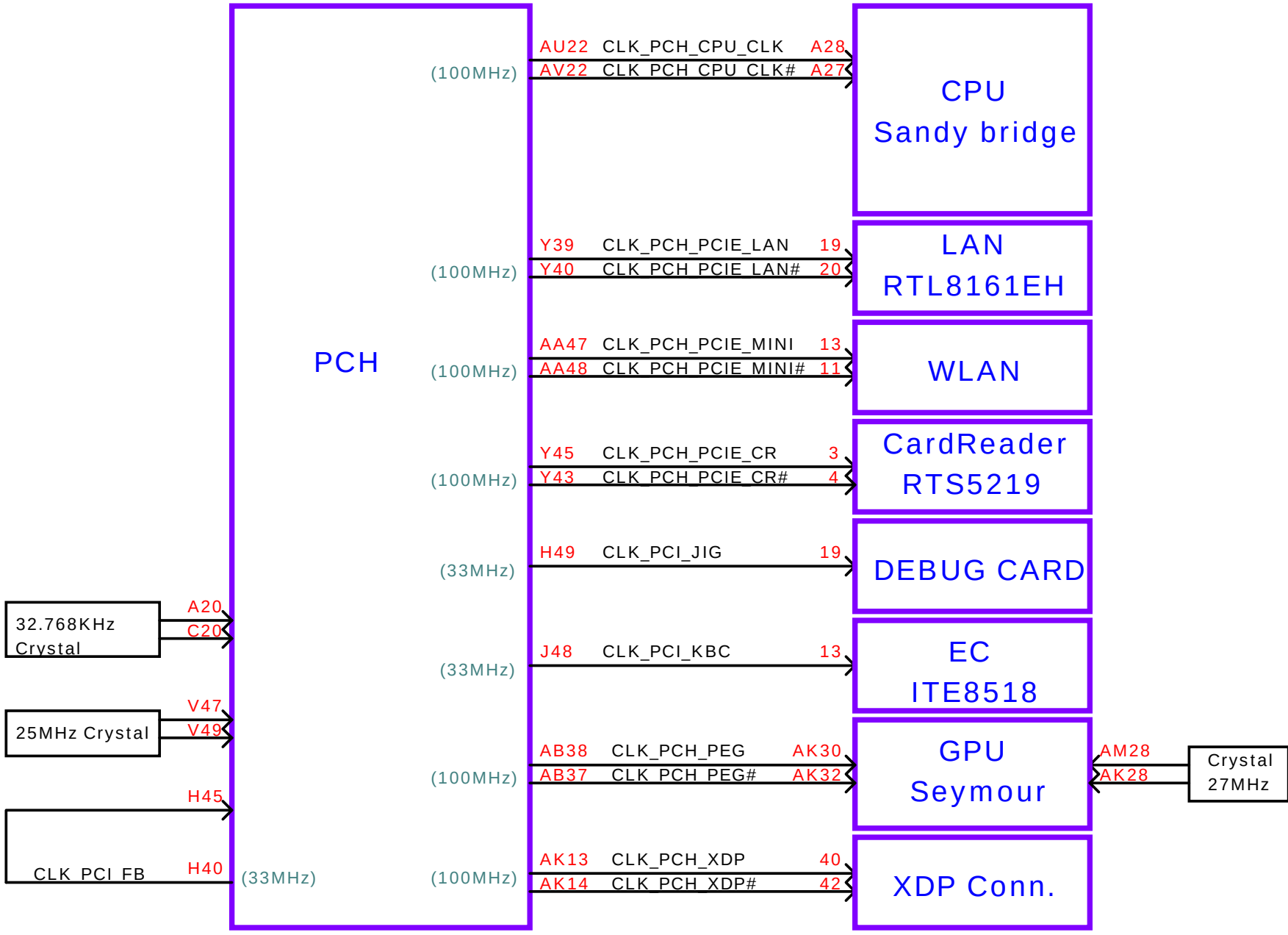
PROJECT : CHICAGO (For Intel Huron River Platform)

01 -- COVER SHEET	23 -- CougarPoint (HDA, SATA..)
02 -- SYSTEM BLOCK DIAGRAM	24 -- CougarPoint (PCI-E, CLK..)
03 -- CLOCK MAP	25 -- CougarPoint (DMI, FDI..)
04 -- POWER SEQUENCY DIAGRAM	26 -- CougarPoint (USB, GPIO..)
05 -- POWER MAP	27 -- CougarPoint (PWR/GND)
06 -- SMBUS MAP	28 -- CougarPoint (PWR, GND)
07 -- Blank	29 -- DDR3 (SO-DIMM 0&1)
08 -- DCIN/BATT	30 -- VGA (PCI-E/STRAP) 1/3
09 -- PWR_CHARGE	31 -- VGA_S3 (IO) 2/3
10 -- PWR_5V/3.3V	32 -- VGA_S3 (DDR3) 3/3
11 -- PWR_VCCP	33 -- VRAM (DDR3)
12 -- PWR_1.5V/0.75S	34 -- EC+KBC (IT8518) & ROM
13 -- PWR_VCORE	35 -- Audio (CODEC_ALC270A)
14 -- PWR_OTHER	36 -- Audio (JACK+AMP+SPK+Mute)
15 -- PWR_ATVDD	37 -- LAN (RTL8161EH)
16 -- PWR_1.8VS	38 -- Mini PCIe & FAN
17 -- PWR_VCCSA	39 -- USBx2/USB DB/SATA CONN.
18 -- Sandy Bridge (DMI, PEG, FDI)	40 -- Card Reader (RTL5219-GR)
19 -- Sandy Bridge (CLK, JTAG..)	41 -- HDMI & CRT
20 -- Sandy Bridge (DDR3)	42 -- LVDS & Webcam
21 -- Sandy Bridge (PWR/GND)	43 -- Sequence circuit
22 -- Sandy Bridge (GRAPHIC PWR)	

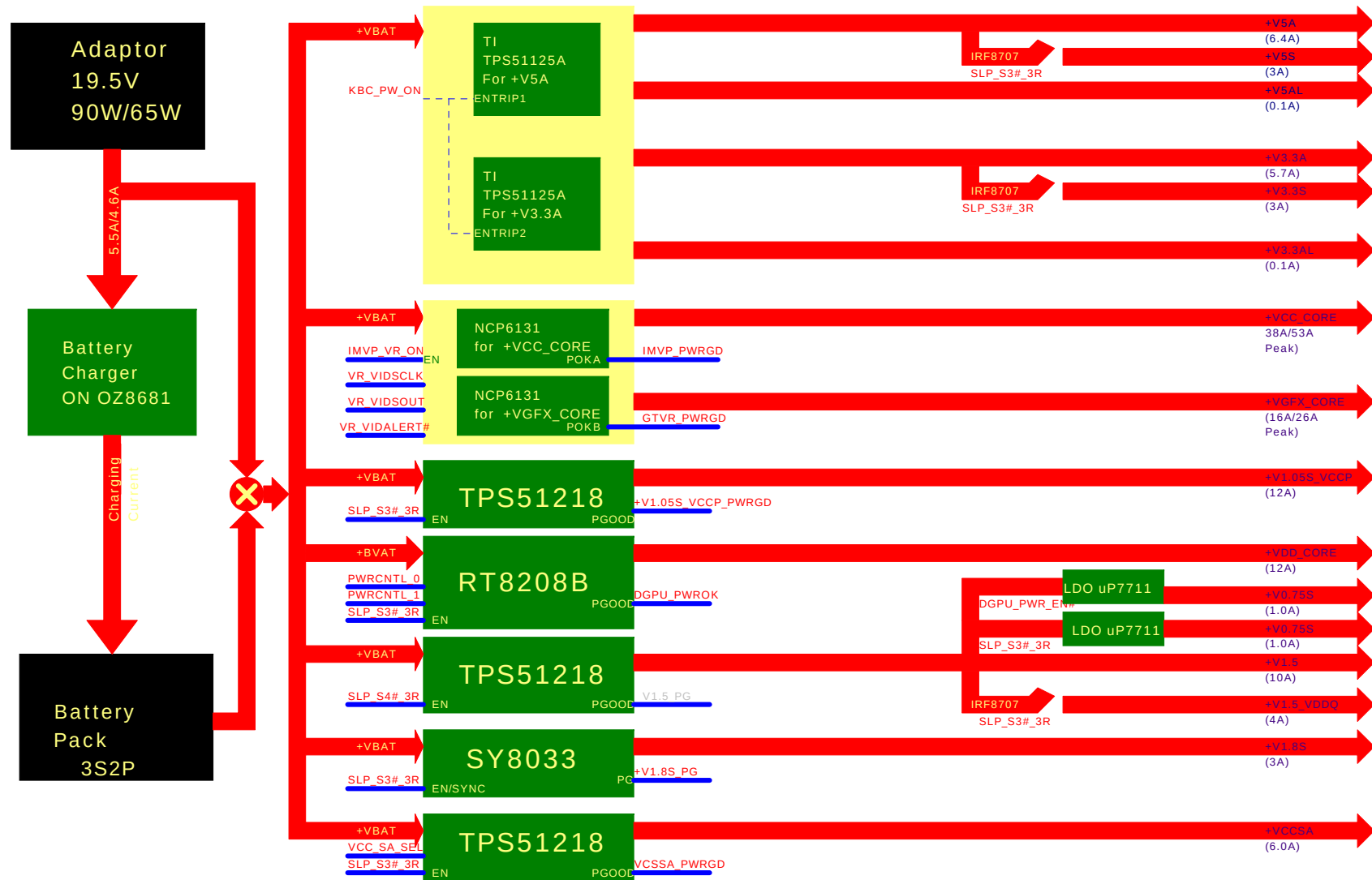
P. Leader	Check by	Design by

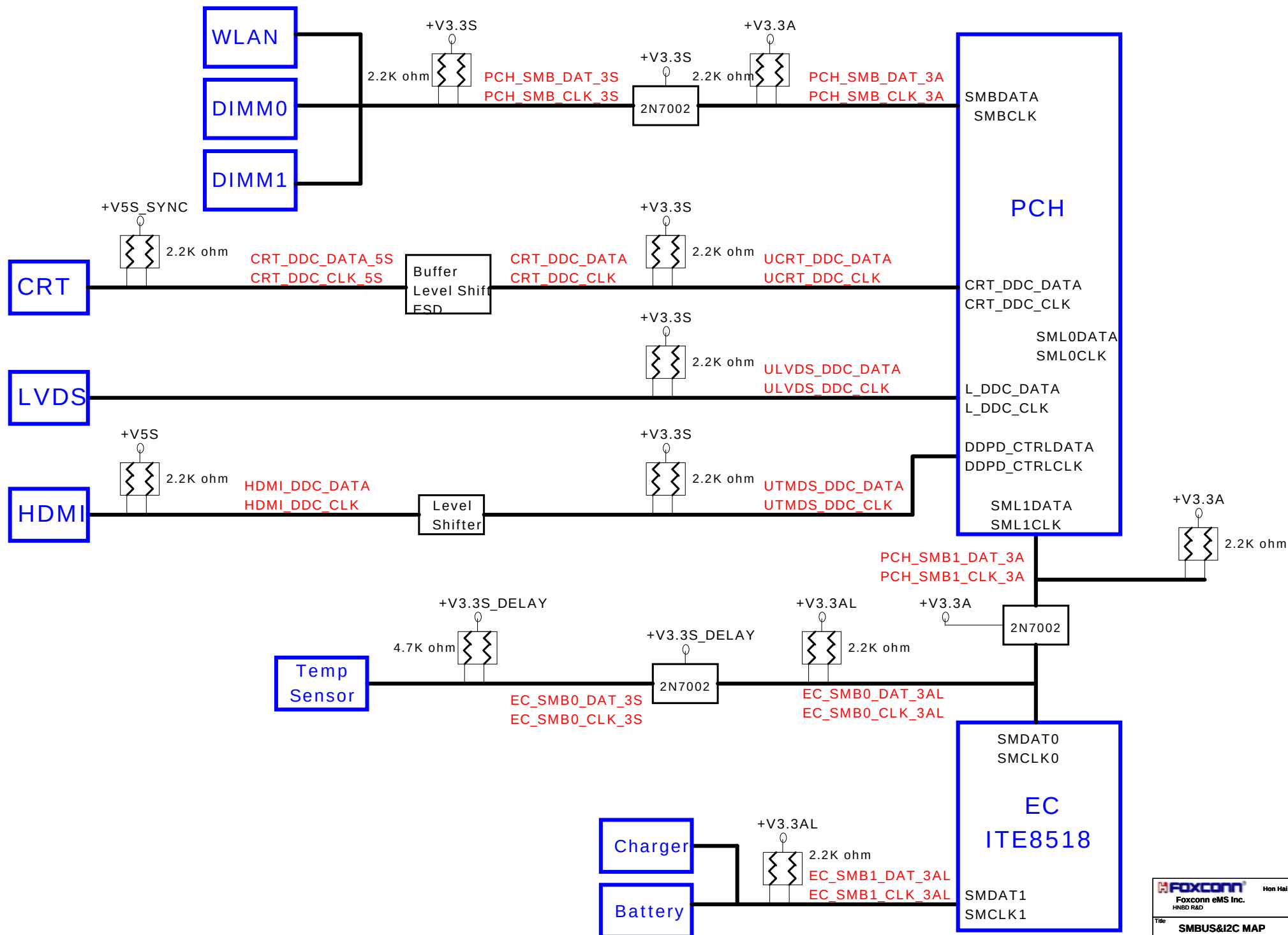
 Foxconn eMS Inc. HNBD R&D		Hon Hai Precision Industry Co. Ltd. phone: +886-2-2799-6111
Title		
Index Page		
Size	Document Number	Rev
Custom	CHICAGO	MV
Page Modified: Tuesday, March 08, 2011		08:28:58 (UTC+GMT) Sheet 1 of 43





POWER MAP

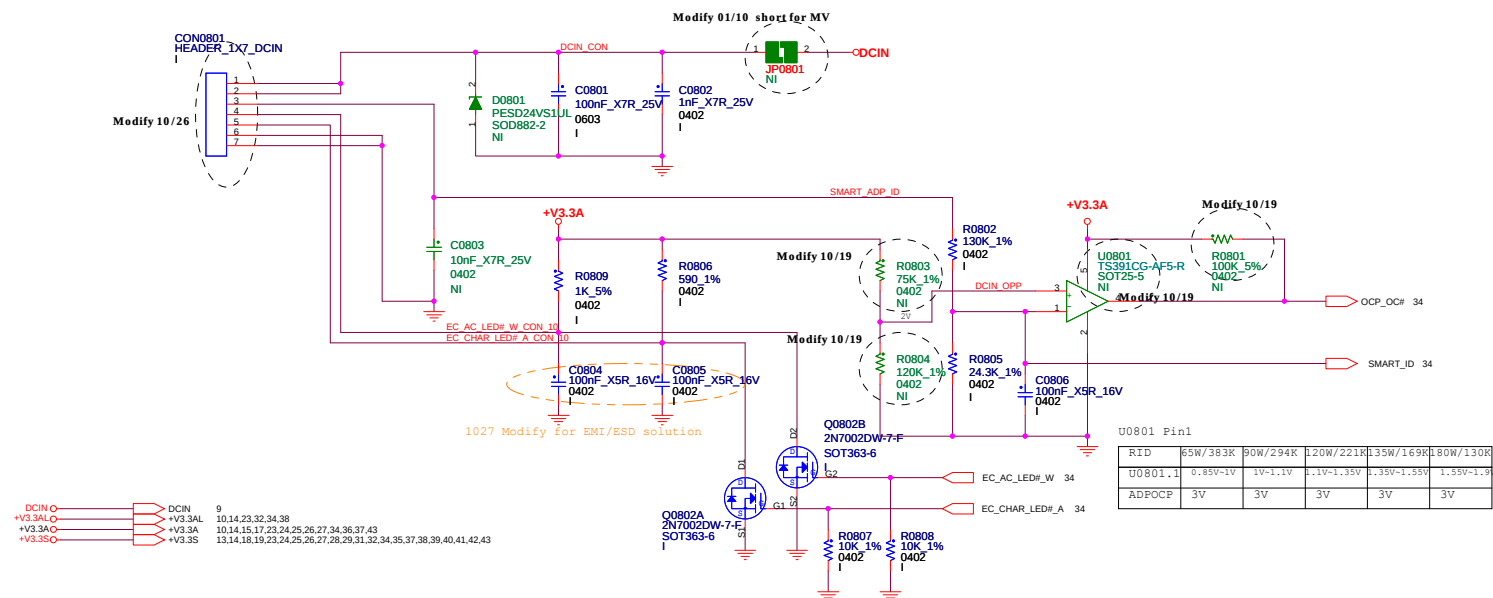




	5	4	3	2	1
D					D
C					C
B					B
A					A
	5	4	3	2	1

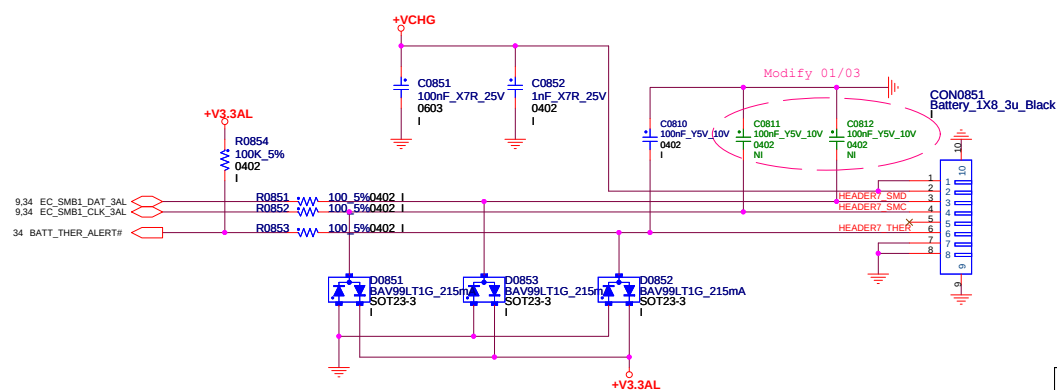
 Hon Hai Precision Industry Co. Ltd. Foxconn eMS Inc. HNBD R&D		phone: +886-2-2799-6111
Title BLANK		
Size A	Document Number CHICAGO	Rev MV
Page Modified: Tuesday, March 08, 2011 08:28:58 (UTC/GMT) Sheet 7 of 43		

2010.1203.0



BATTERY CONNECTOR

2010.0914.0



BATTERY CHARGER

TRAVELADP	DCIN RANGE	AC_OFF	CHARGE
>2.412V	>17V	LOW	YES
1.844V<->2.412V	13V<->17V	LOW	NO
<1.844V	<13V	HIGH	NO

8.34 EC_SMB1_CLK_3AL

8.34 EC_SMB1_DAT_3AL

DCIN 8
+VBAT 10.11.12.13.15.42.43
+VCHG 8
+V3.3AL0 8.10.14.23.32.34.38

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HNB0 RAD
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phone: +886-2-2799-6111

Title
PWR_Charger
Size
Custom
Document Number
CHICAGO
Rev
MV
Page Modified: Tuesday, March 06, 2012 09:28:58 (UTC+0800) Sheet 9 of 43

TRAVELADP	DCIN_RANGE	AC_OFF	CHARGE
>2.412V	>17V	LOW	YES
1.844V<=>2.412V	13V<=>17V	LOW	NO
<1.844V	<13V	HIGH	NO

8,34 EC_SMB1_CLK_3AL

8,34 EC_SMB1_DAT_3AL

DCIN	8
+VBAT	10,11,12,13,15,42,43
+VCHG	8
+V3.3AL	8,10,14,23,32,34,38

2010.1103.0

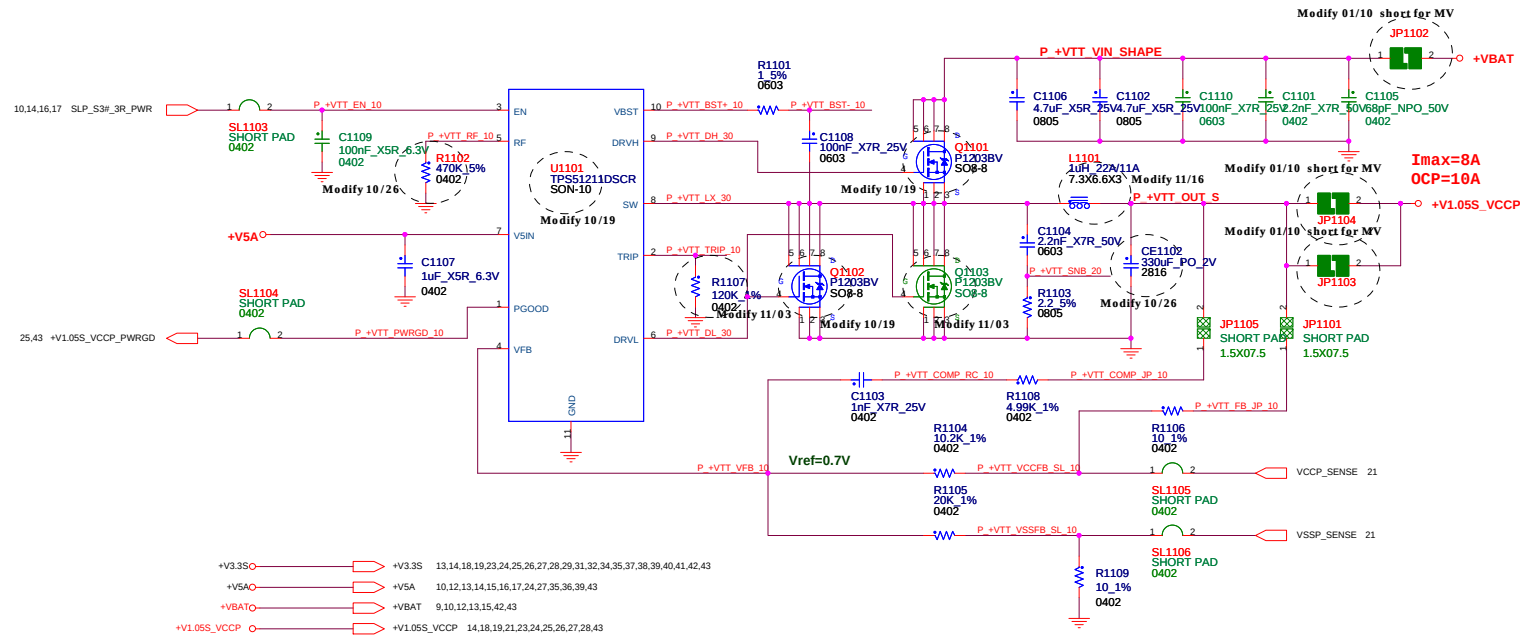


2010.1103.0

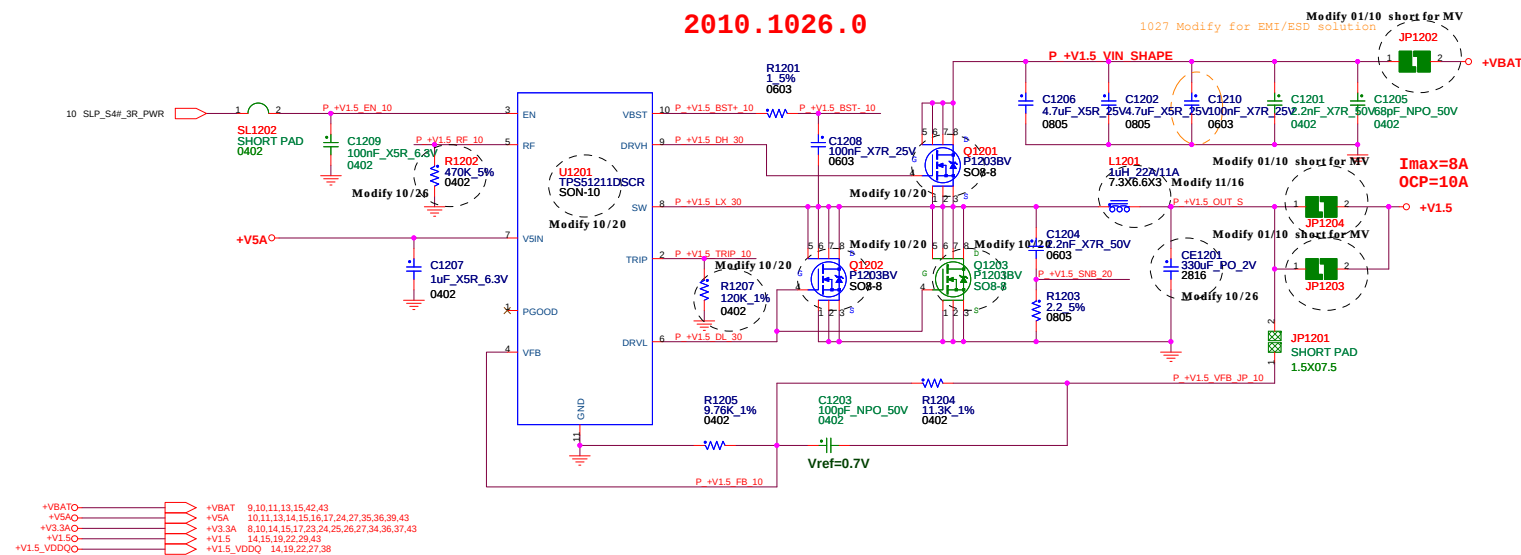
+V1.05S_VCCP:

1. I/P Current:
 $I_{in} = V_o \cdot I_o / (0.75 \cdot V_{in}) = 1.24A$
2. Ripple Current:
 $I_{rip} = 3.42A$
3. Ripple Voltage:
 $ESR/1 = 9m\Omega$
 $V_{rip} = 30.78mV$
4. Inductor Spec:
 $I_{sat} = 36A$
 $I_{dc} = 18A$
 $DCR = 3.3m\Omega$
5. MOSFET Spec:

H-side MOSFET: IRF8707PBF	L-side MOSFET: IRF8707PBF
$R_{ds(ON)} = 17.5m\Omega$ ($V_{gs} = 4.5V$)	$R_{ds(ON)} = 17.5m\Omega$ ($V_{gs} = 4.5V$)
$I_{cont} = 11A$ ($T = 25^\circ C$)	$I_{cont} = 11A$ ($T = 25^\circ C$)
$I_{peak} = 88A$ (Pause = 10 us)	$I_{peak} = 88A$ (Pause = 10 us)
6. Frequency:
 $F = 290KHz$ ($R_{1102} = 0\Omega$)
7. OCP:
 $Set = R_{1107}$ to 120K
 $V_{trip} = R_{1107} \cdot 10uA = 1.2V$
 $locp = (V_{trip} / 8 \cdot R_{ds(on)}) + I_{ripple} / 2 = 10A$

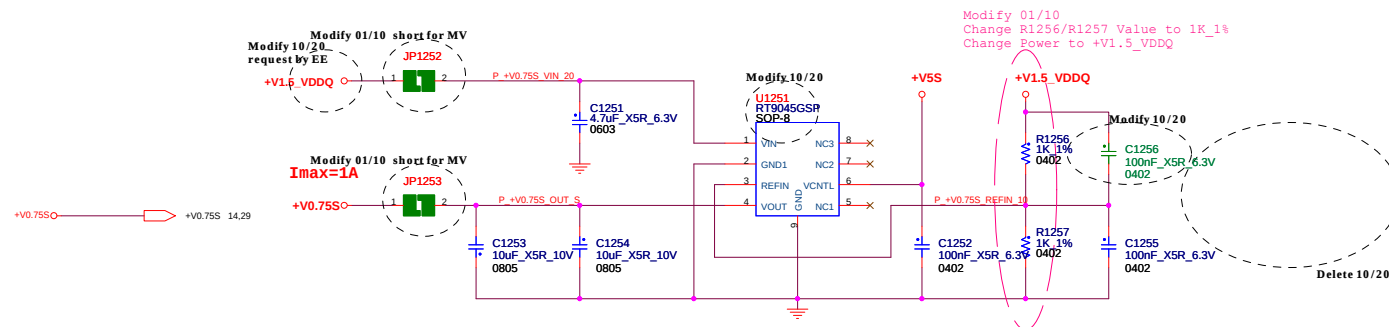


2010.1026.0



- +V1.5:
- 1. I/P Current:
 $I_{in} = V_o \cdot I_o / (0.75 \cdot V_{in}) = 1.78A$
 - 2. Ripple Current:
 $I_{rip} = 3.34A$
 - 3. Ripple Voltage:
 $ESR/1 = 9m\Omega$
 $V_{rip} = 30.6mV$
 - 4. Inductor Spec:
 $I_{sat} = 36A$
 $I_{dc} = 18A$
 $DCR = 3.3m\Omega$
 - 5. MOSFET Spec:
H-side MOSFET: IRF8707PBF
 $R_{ds(ON)} = 17.5m\Omega$ ($V_{gs} = 4.5V$)
 $I_{cont} = 11A$ ($T = 25^\circ C$)
 $I_{peak} = 88A$ (Pause = 10 us)
L-side MOSFET: IRF8707PBF
 $R_{ds(ON)} = 17.5m\Omega$ ($V_{gs} = 4.5V$)
 $I_{cont} = 11A$ ($T = 25^\circ C$)
 $I_{peak} = 88A$ (Pause = 10 us)
 - 6. Frequency:
 $F = 290KHz$ ($R_{0902} = 0\Omega$)
 - 7. OCP:
Set = R1207 to 120K
 $V_{trip} = R1207 \cdot 10\mu A = 1.2V$
 $I_{ocp} = (V_{trip} / 8 \cdot R_{ds(on)}) + I_{ripple} / 2 = 10A$

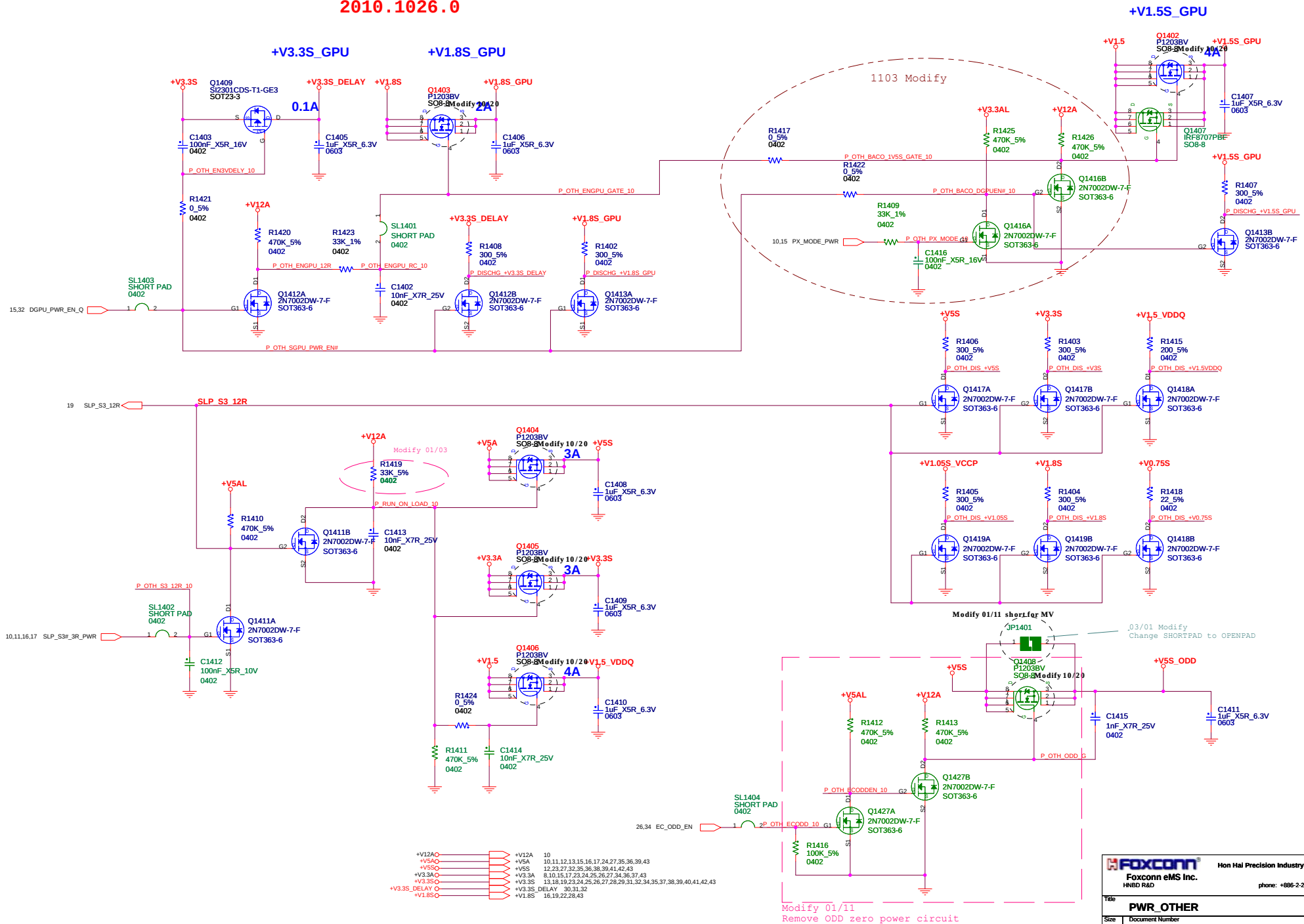
2010.1026.0



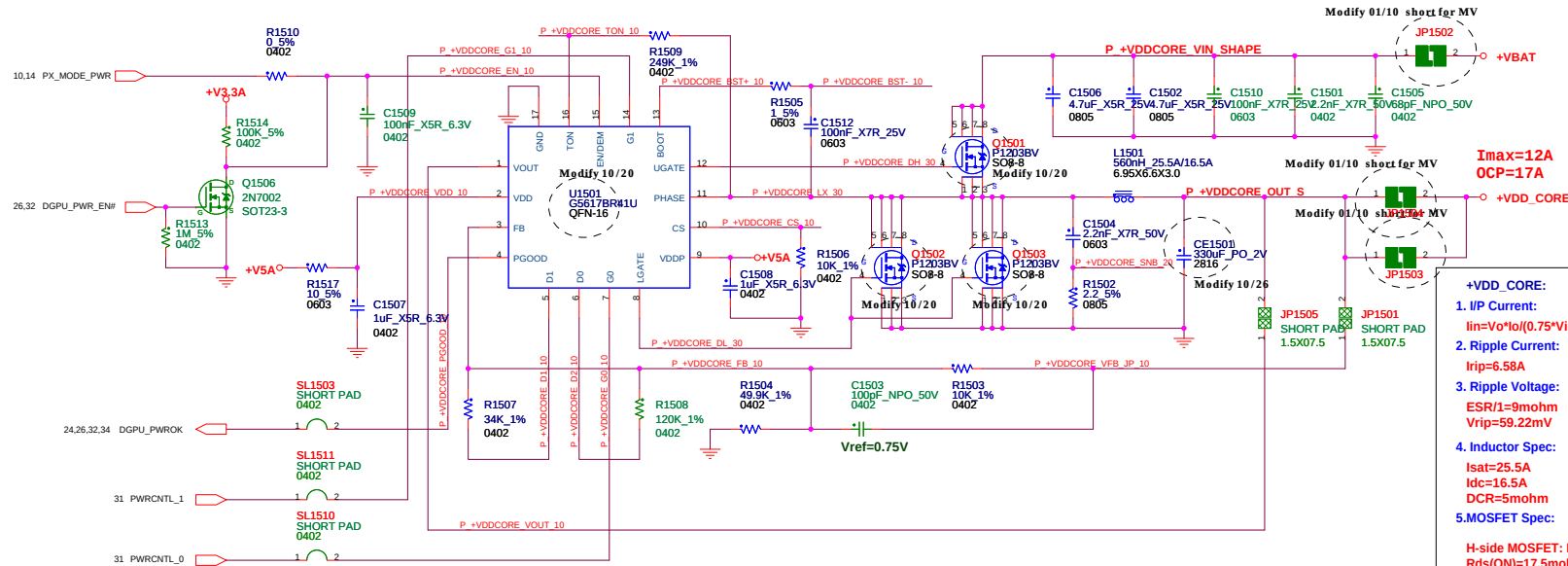
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2010.1026.0



2010.1026.0

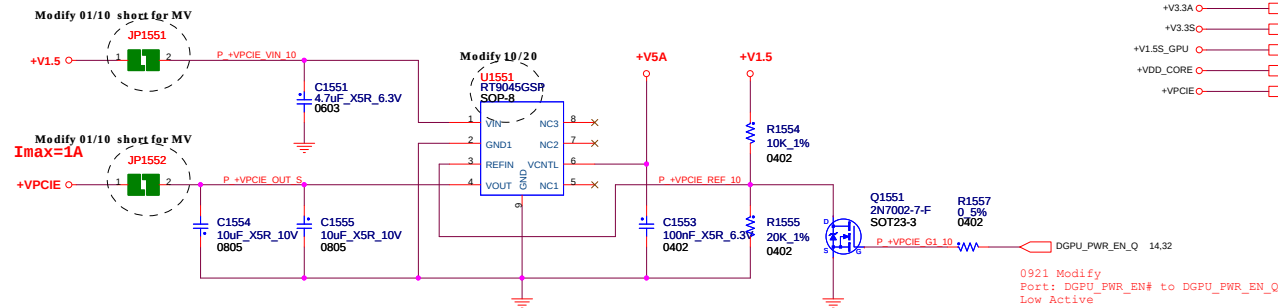


PWRCNTL_1	PWRCNTL_0	VDD_CORE
0	---	1.121V
---	---	---
1	---	0.9V
---	---	---

+VDD_CORE:

1. **IIP Current:**
 $I_{in} = V_o * I_o / (0.75 * V_{in}) = 1.48A$
2. **Ripple Current:**
 $I_{rip} = 6.58A$
3. **Ripple Voltage:**
 $ESR / I_1 = 9m\Omega$
 $V_{rip} = 59.22mV$
4. **Inductor Spec:**
 $I_{sat} = 25.5A$
 $I_{dc} = 16.5A$
 $DCR = 5m\Omega$
5. **MOSFET Spec:**
- | | |
|--|--|
| H-side MOSFET: IRF8707PBF | L-side MOSFET: IRF8707PBF |
| $R_{ds(ON)} = 17.5m\Omega$ ($V_{gs} = 4.5V$) | $R_{ds(ON)} = 17.5m\Omega$ ($V_{gs} = 4.5V$) |
| $I_{cont} = 11A$ ($T = 25^\circ C$) | $I_{cont} = 11A$ ($T = 25^\circ C$) |
| $I_{peak} = 88A$ (Pause $\geq 10us$) | $I_{peak} = 88A$ (Pause $\geq 10us$) |
6. **Frequency:**
 $TON = 9.6P * R_{1509} * (V_{OUT} + 0.1) / (V_{IN} - 0.3) + 50ns = 206ns$
 $F = V_{OUT} / (V_{IN} * TON) = 286KHz$
7. **OCp:**
 $Set = R_{1506} \text{ to } 10K$
 $V_{trip} = R_{1206} * I_{0uA} = 0.1V$
 $l_{ocp} = (V_{trip} / R_{dson}) + I_{ripple} / 2 = 17A$

2010.1020.0 +VPCIE POWER SUPPLY



+VBAT		+VBAT	9.10.11.12.13.42.43
+V5A		+V5A	10.11.12.13.14.16.17.24.27.35.36.39.43
+V3.3A		+V3.3A	8.10.14.17.23.24.25.26.27.34.36.37.43
+V3.3S		+V3.3S	13.14.18.19.23.24.25.26.27.28.29.31.32.34.35.37.38.39.40.41.42.43
+V1.55_GPU		+V1.55_GPU	14.30.32.33.43
+VDD_CORE		+VDD_CORE	32.43
+VPCIE		+VPCIE	30.31.32.43

+V1.8S POWER SUPPLY

2010.1025.0

+V1.8S:

1. I/P Current:

$$I_{in} = V_o * I_o / (0.75 * V_{in}) = 1.44A$$

2. Ripple Current:

$$I_{rip} = 0.53A$$

3. Ripple Voltage:

$$ESR/3 = 3.3m\Omega$$

$$V_{rip} = 1.75mV$$

4. Inductor Spec:

$$I_{sat} = 14A$$

$$I_{dc} = 8A$$

$$DCR = 20m\Omega$$

5. MOSFET Spec:

H-side P-MOSFET:

L-side N-MOSFET:

$$R_{ds(ON)} = 110m\Omega \quad (V_{gs} = 4.5V)$$

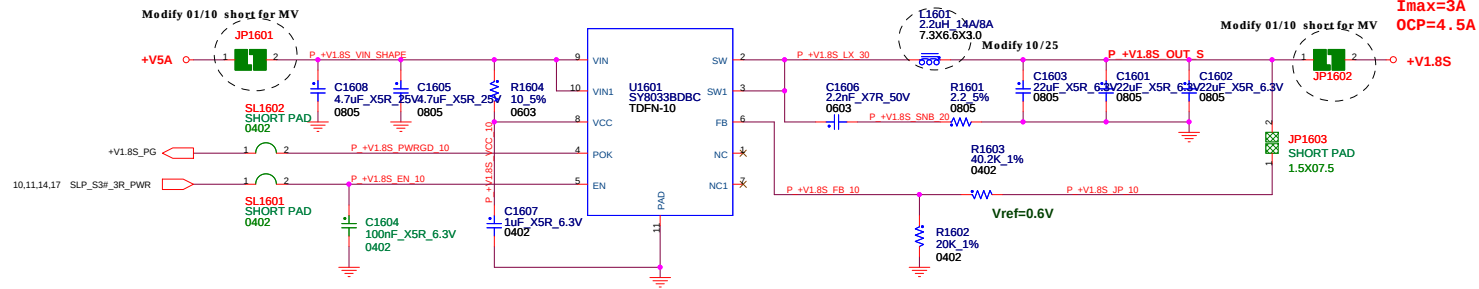
$$R_{ds(ON)} = 75m\Omega \quad (V_{gs} = 4.5V)$$

6. Frequency:

$$F = 1MHz \quad (\min = 800KHz, \max = 1.2MHz)$$

7. OCP:

$$I_{ocp} = 4A(\min) / 4.5A(\text{typ}) / 5A(\max)$$



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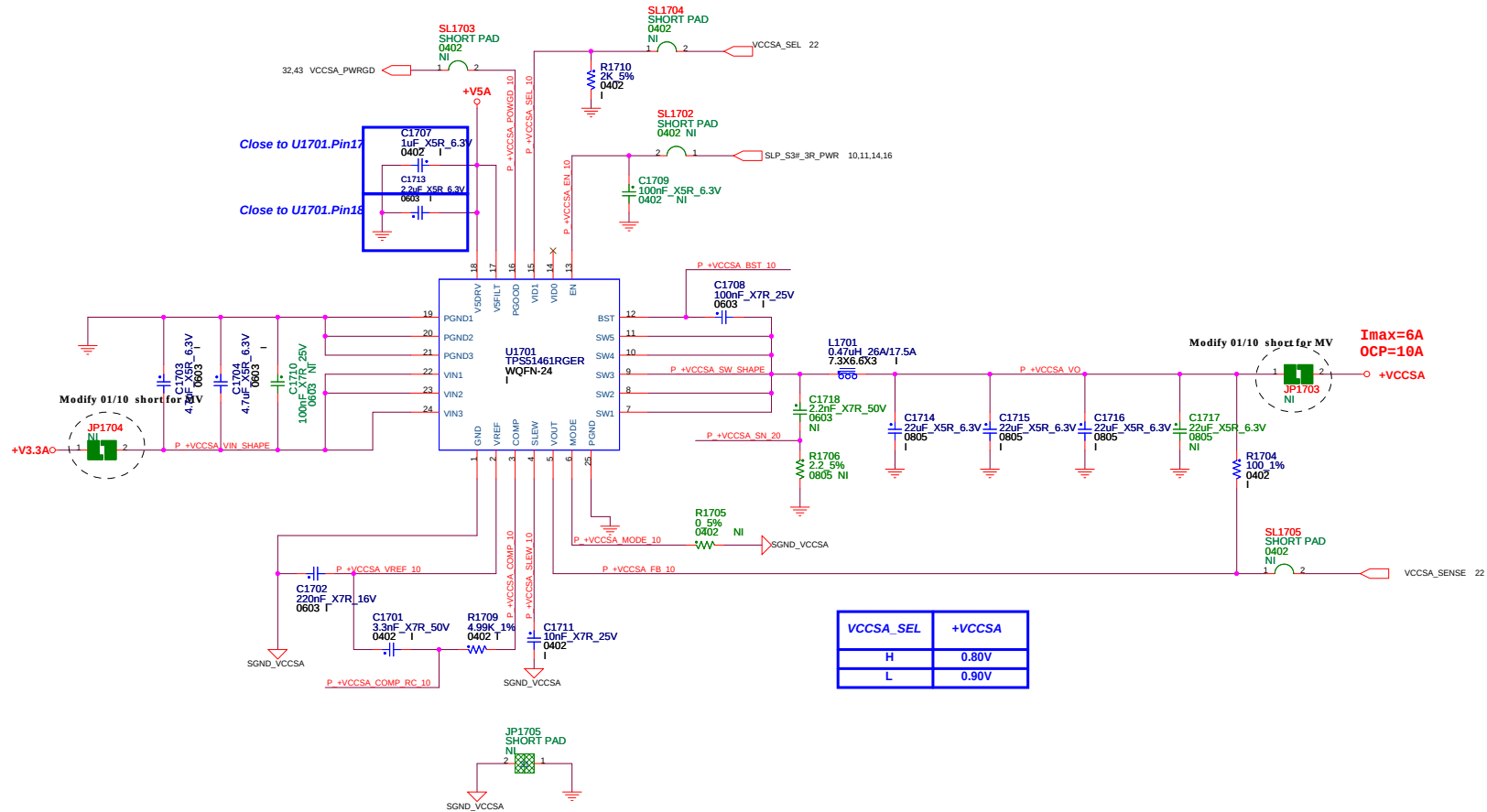
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Title		+1.8VS
Size	Document Number	
C	CHICAGO	Rev
Page Modified: Tuesday, March 08, 2011		062800_01C00001
Sheet 16		of 43

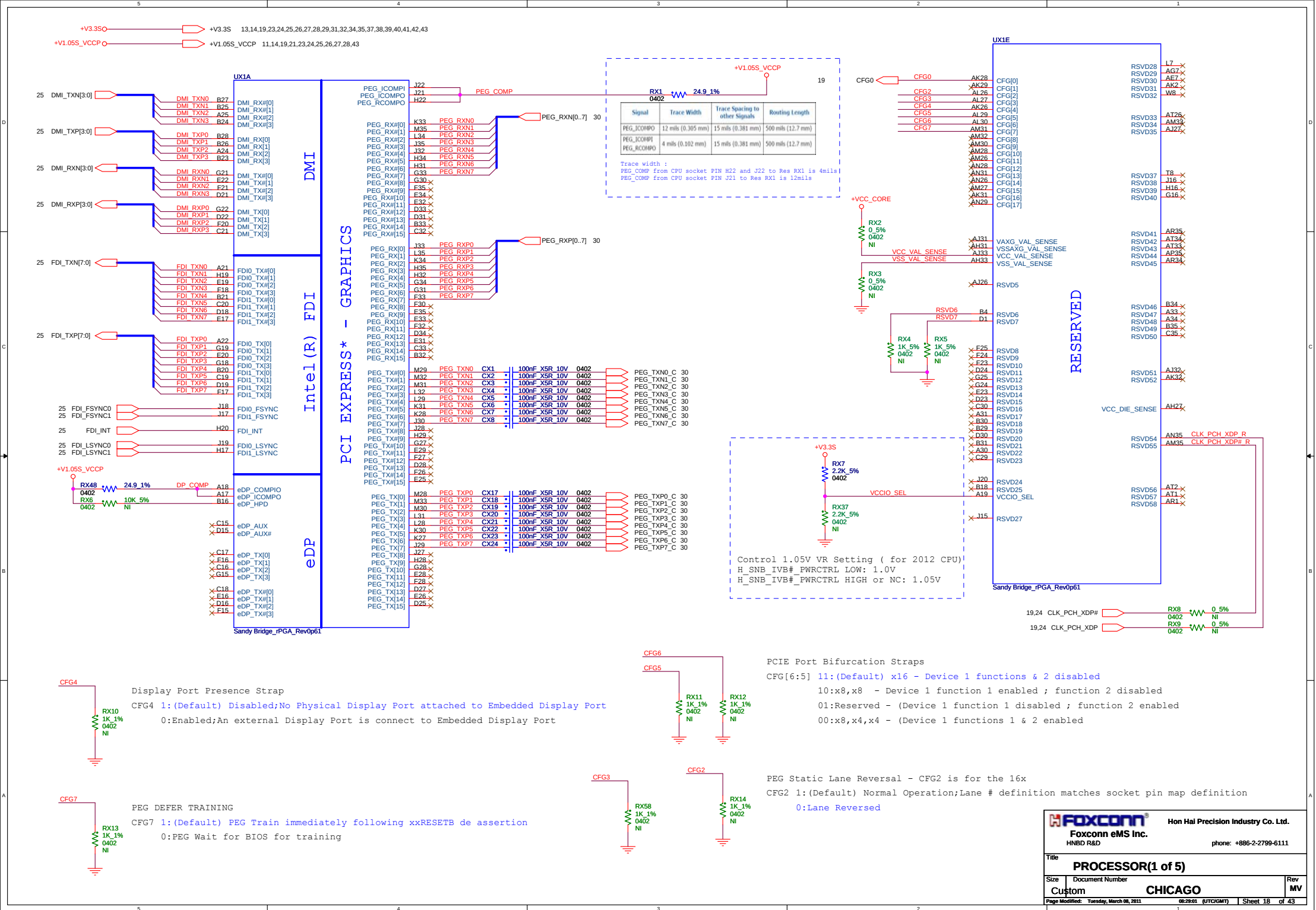
+VCCSA POWER SUPPLY

2010.1026.0

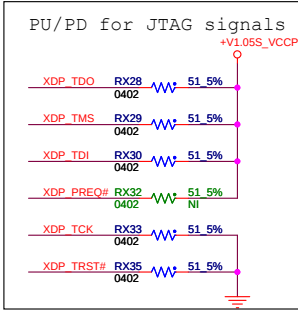
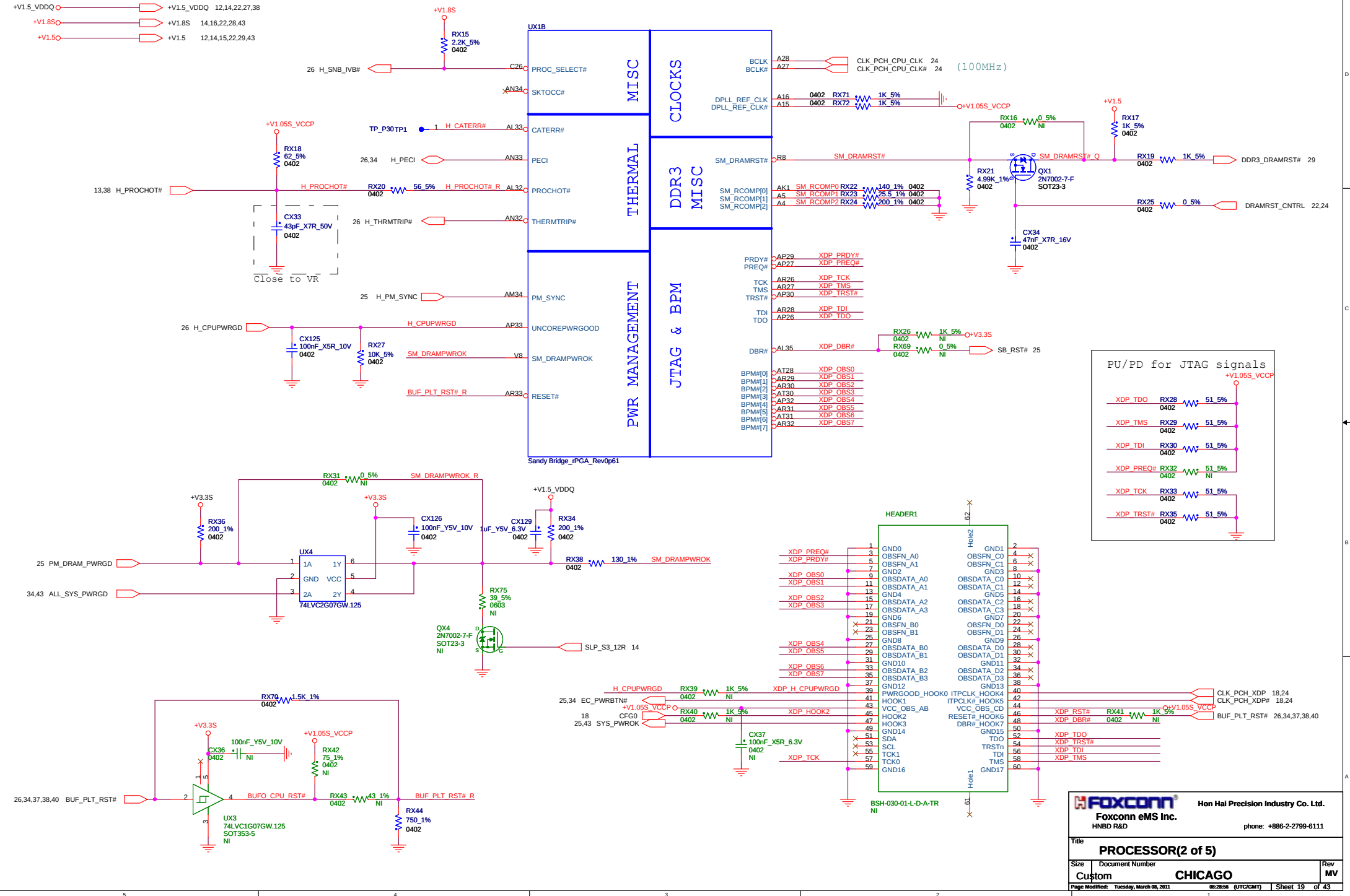


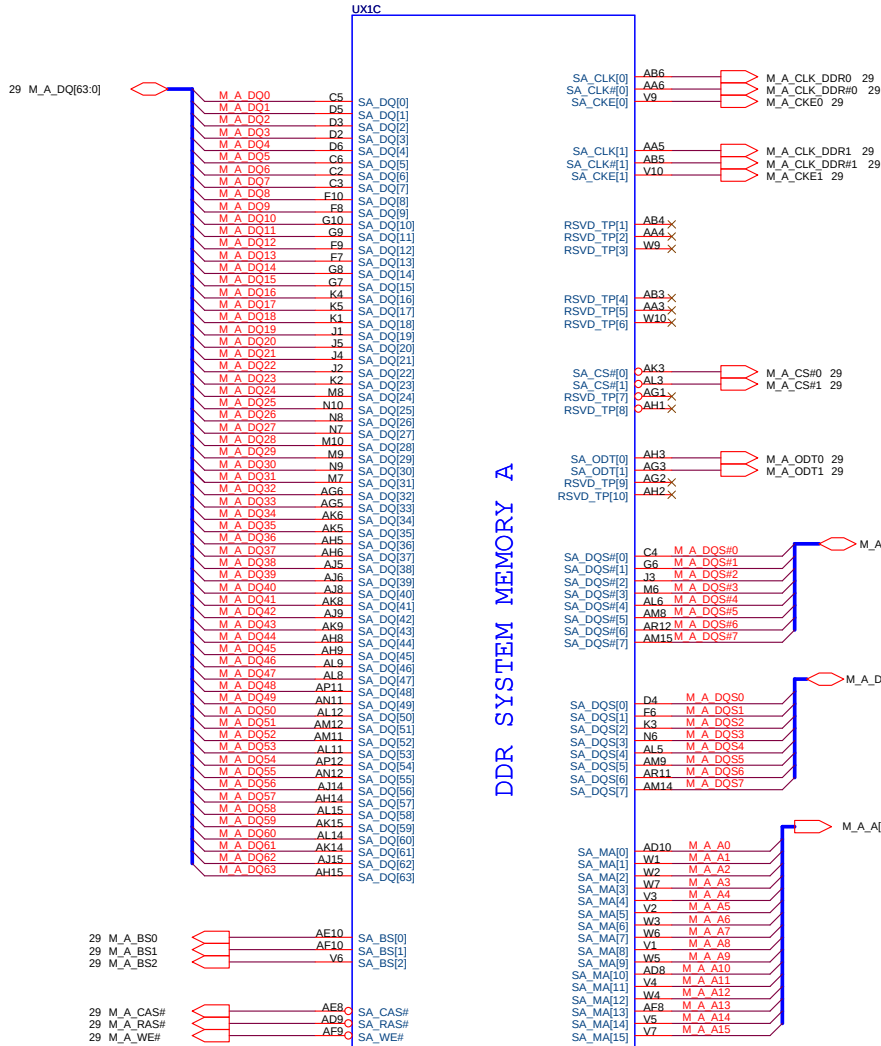
- +VCCSA:**
- I/P Current:**
 $I_{in} = V_o \cdot I_o / (0.75 \cdot V_{in}) = 2.18A$
 - Ripple Current:**
 $I_{rip} = 1.39A$
 - Ripple Voltage:**
 $ESR/4 = 1m\Omega$
 $V_{rip} = 1.39mV$
 - Inductor Spec:**
 $I_{sat} = 26A$
 $I_{dc} = 17.5A$
 $DCR = 4.2m\Omega$
 - MOSFET Spec:**

- Frequency:**
 $F = 1MHz$ (R1705=Open)
- OC:**
Min : 6A / Typ : 7.5A



+V3.3SQ → +V3.3S 13,14,18,23,24,25,26,27,28,29,31,32,34,35,37,38,39,40,41,42,43
+V1.05S_VCCP → +V1.05S_VCCP 11,14,18,21,23,24,25,26,27,28,43
+V1.5_VDDQ → +V1.5_VDDQ 12,14,22,27,38
+V1.8SQ → +V1.8S 14,16,22,28,43
+V1.5C → +V1.5 12,14,15,22,29,43





DDR SYSTEM MEMORY A

Sandy Bridge_rPGA_RevOp61

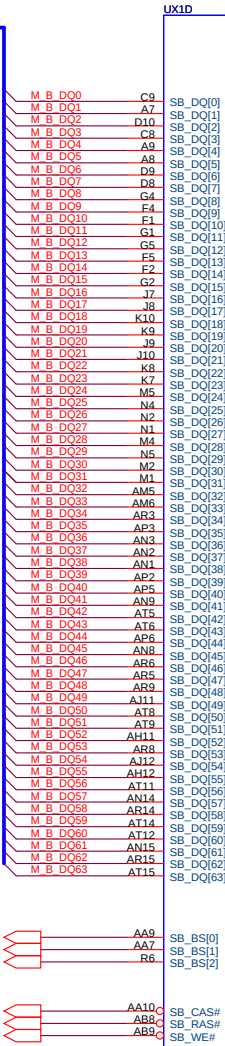
29 M_B_DQ[63:0]

29 M_B_DQ[63:0]

29 M_B_BS0
29 M_B_BS1
29 M_B_BS2

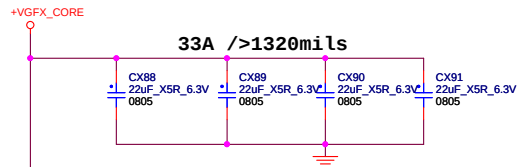
29 M_B_CAS#
29 M_B_RAS#
29 M_B_WE#

29 M_B_CAS#
29 M_B_RAS#
29 M_B_WE#



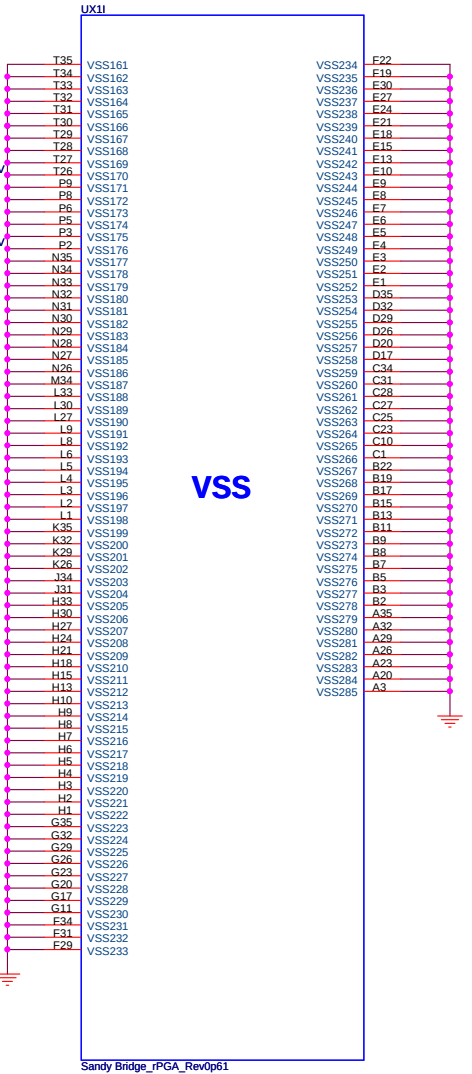
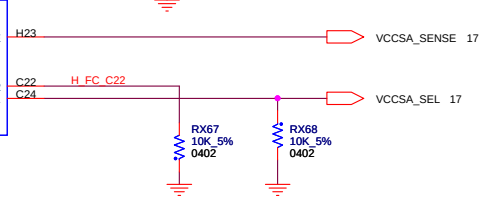
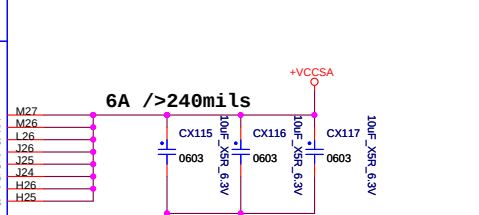
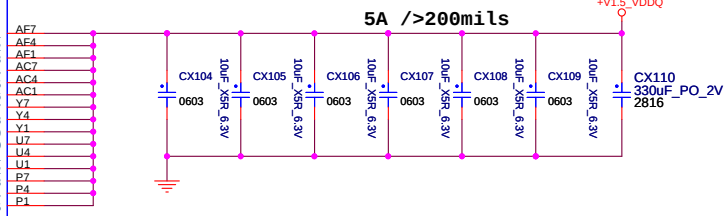
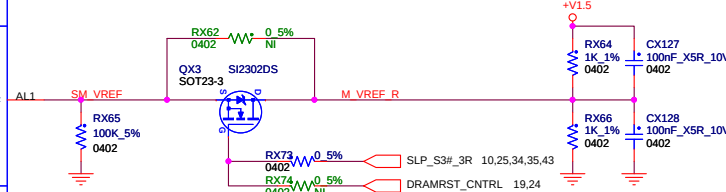
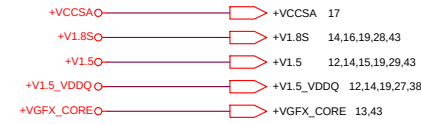
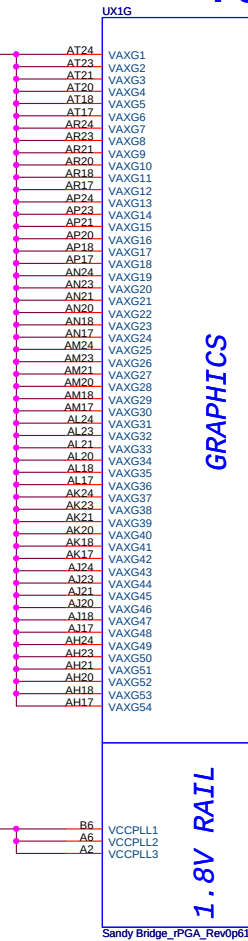
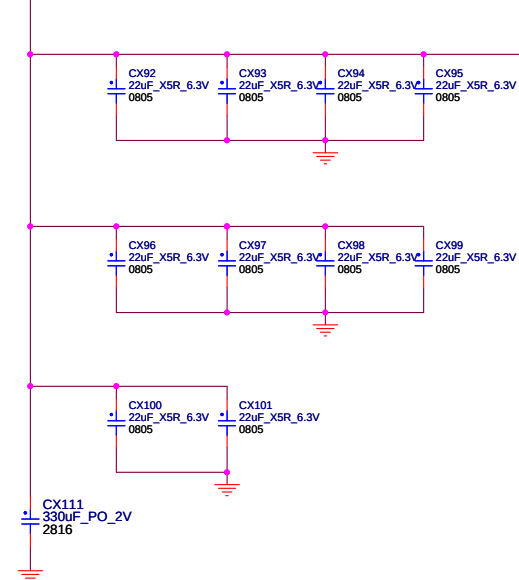
DDR SYSTEM MEMORY B

Sandy Bridge_rPGA_RevOp61

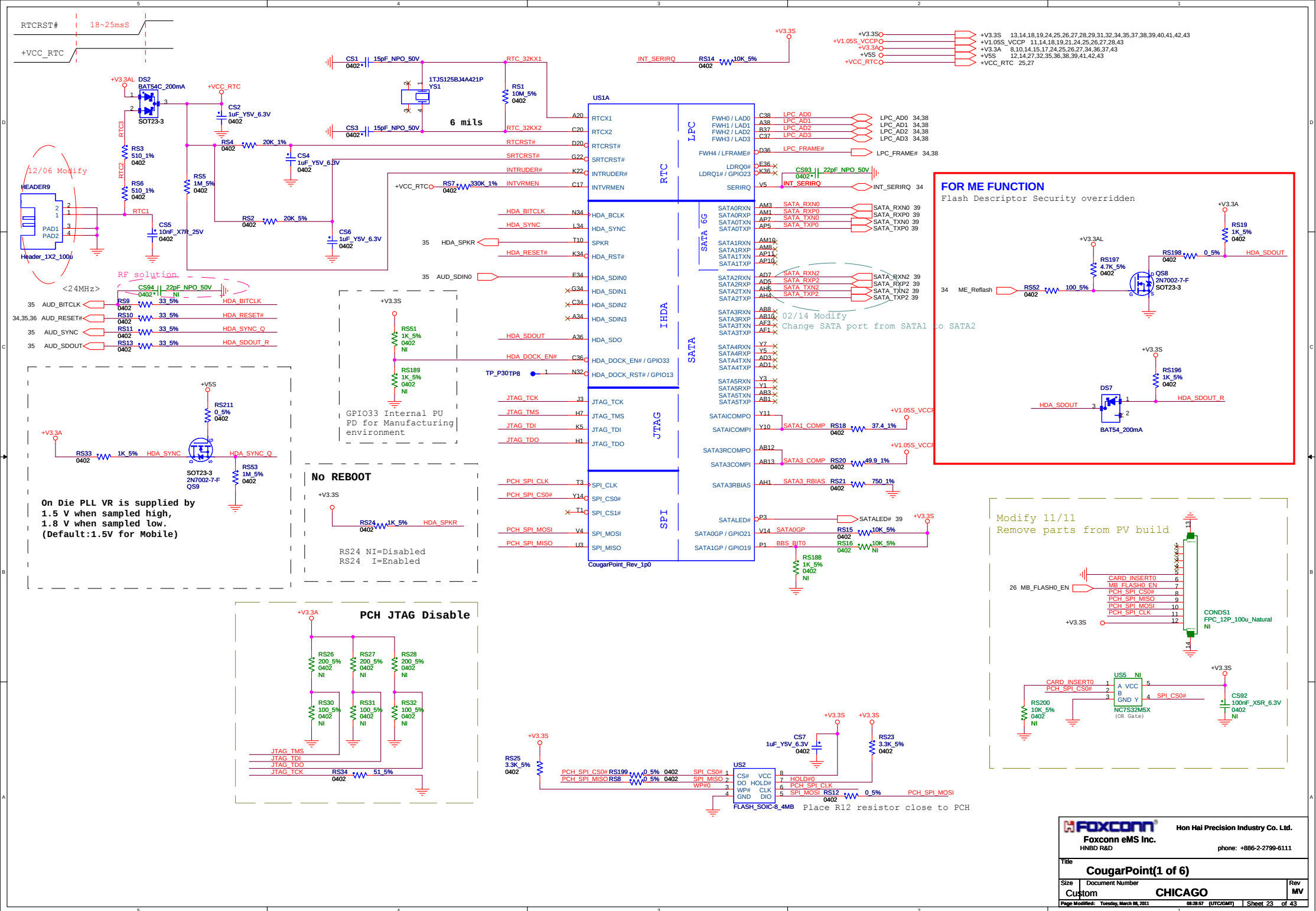


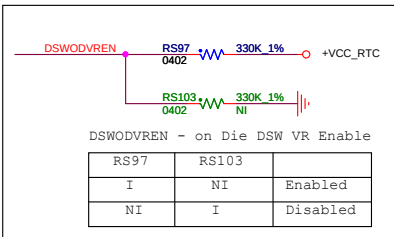
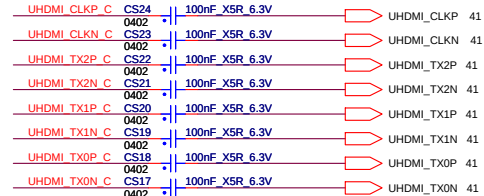
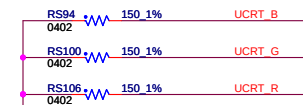
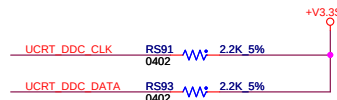
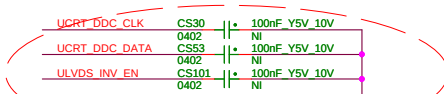
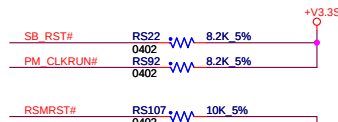
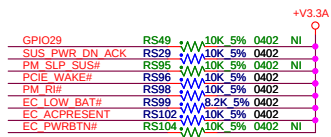
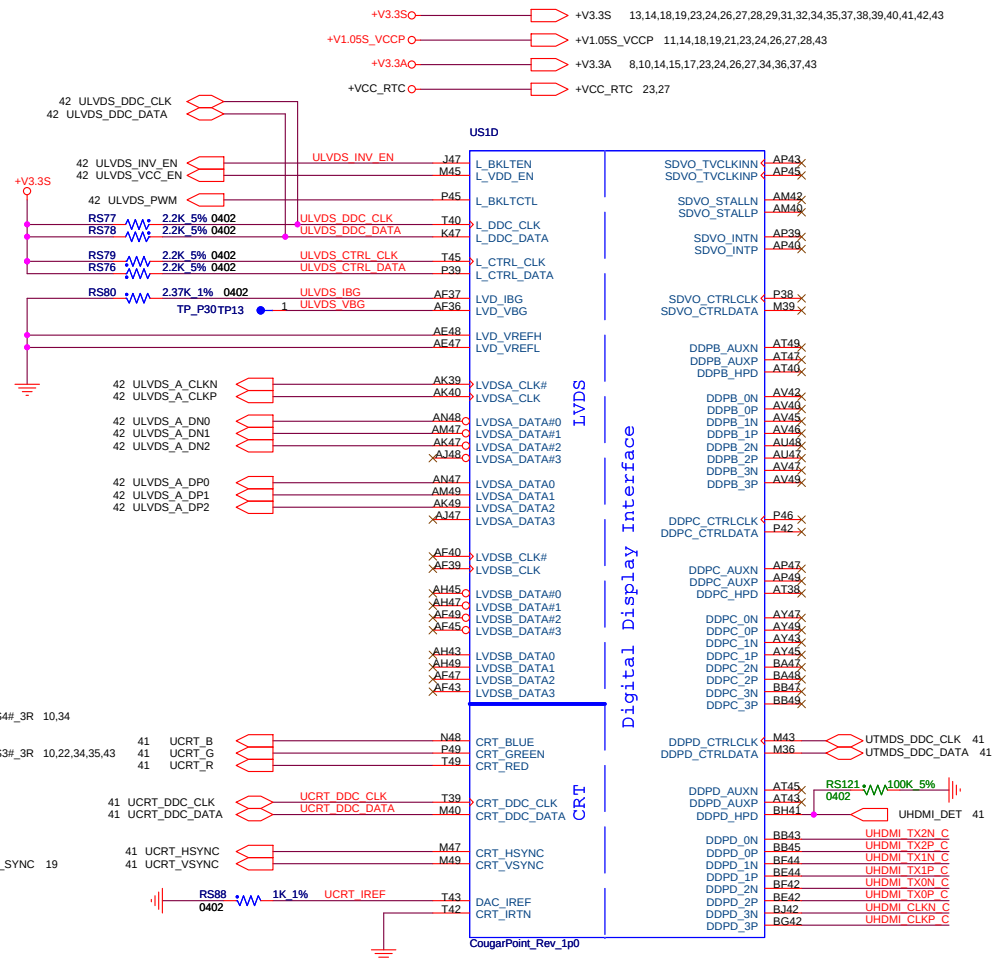
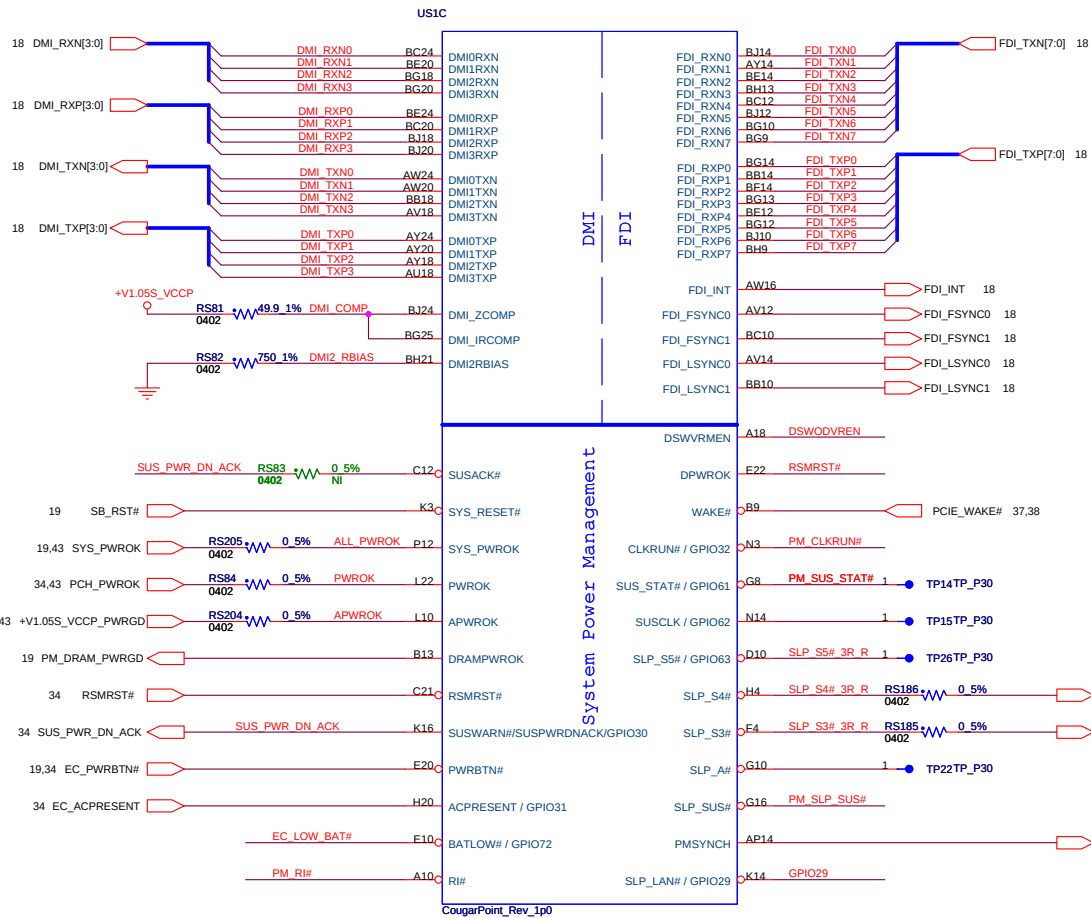
FOR VAXG:
2x 330 μ F Bottom Edge,
4x 0805 22 μ F Top & Bottom Cavity,
8x 0805 22 μ F Top & Bottom Edge,

POWER



FOR VCCPLL:
1x 330 μ F Bottom Edge,
2x 0402 1 μ F Bottom Edge,
1x 0805 10 μ F Bottom Edge,



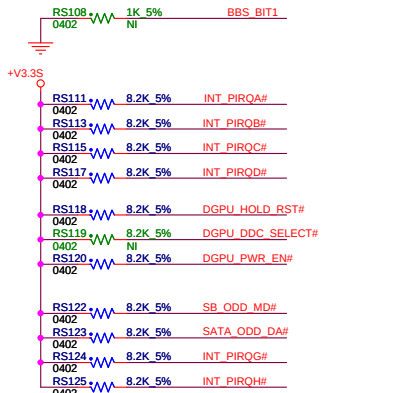


12/06 Modify

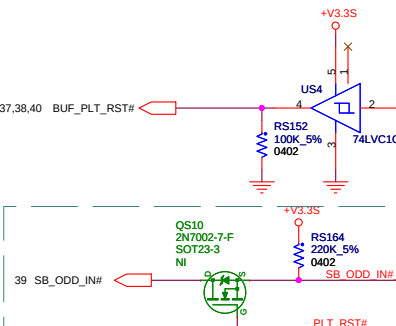
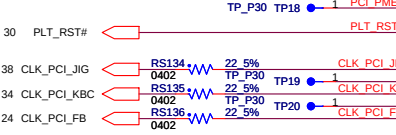
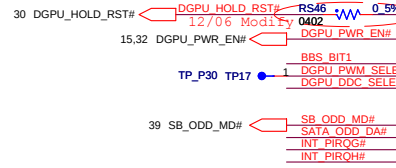
Modify CS30/CS53/CS101 to non-stuff on 01/11

Place resistor close to PCH

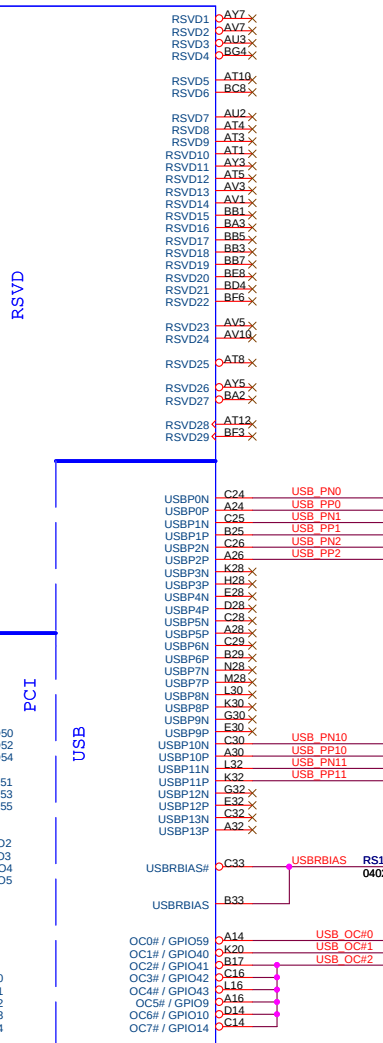
Boot BIOS Strap		
BBS_BIT1	BBS_BIT0	Boot BIOS Location
0	0	LPC
0	1	Reserved (NAND)
1	0	PCI
1	1	SPI



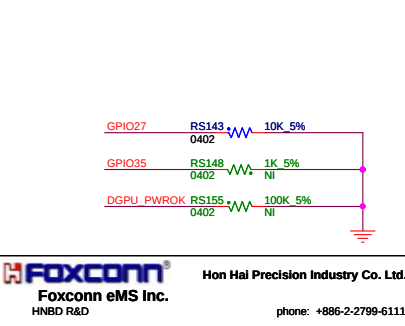
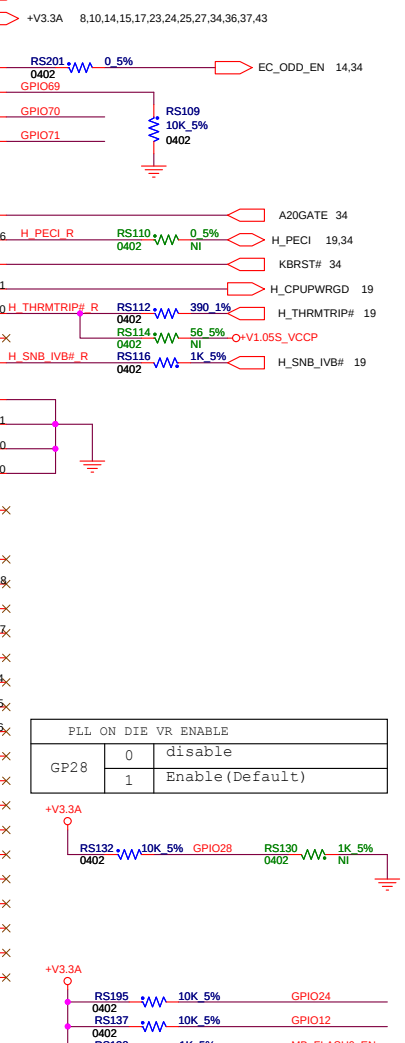
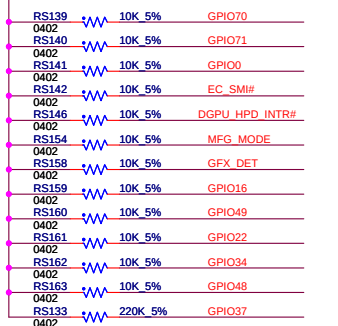
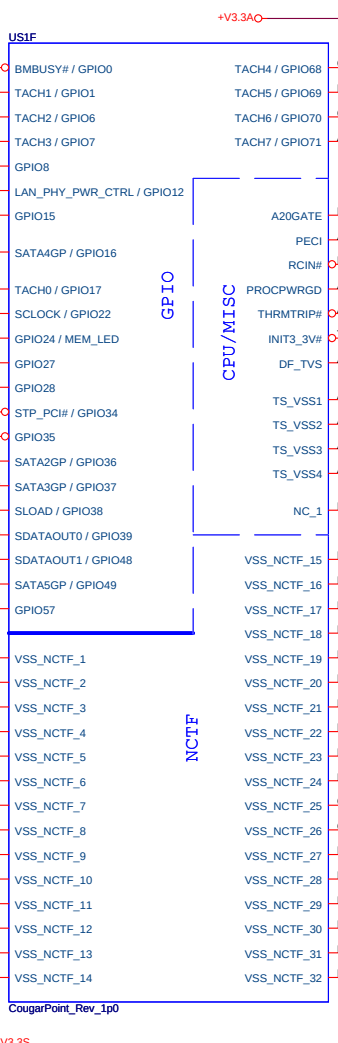
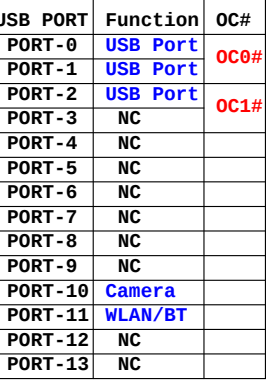
Del RS121 8.2Kohm to +V3.3S
0820 Update



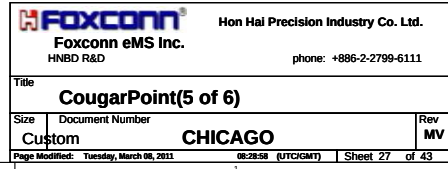
02/24 Modify
same as the PV build



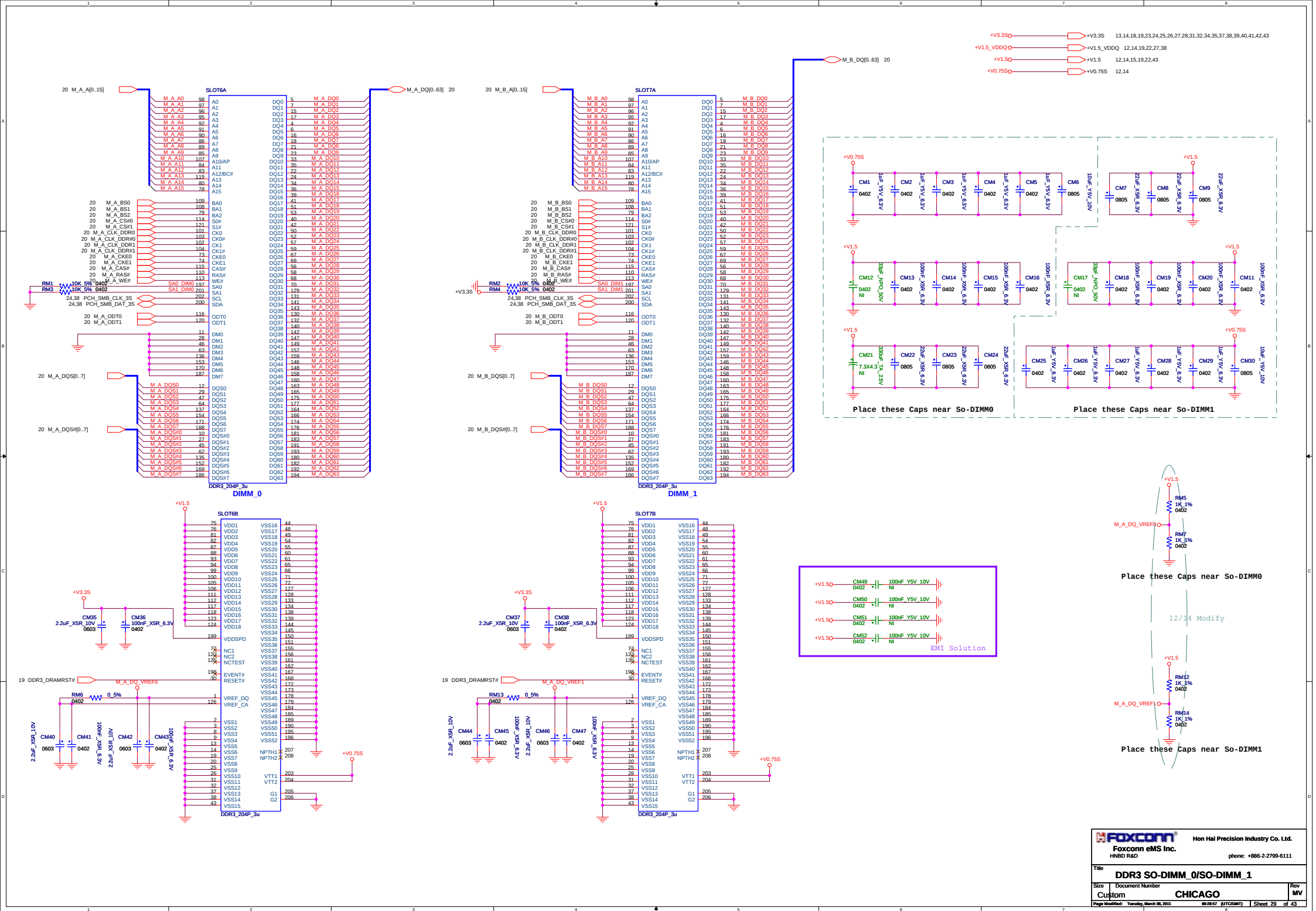
USB PORT	Function	OC#
PORT-0	USB Port	OC0#
PORT-1	USB Port	OC0#
PORT-2	USB Port	OC1#
PORT-3	NC	
PORT-4	NC	
PORT-5	NC	
PORT-6	NC	
PORT-7	NC	
PORT-8	NC	
PORT-9	NC	
PORT-10	Camera	
PORT-11	WLAN/BT	
PORT-12	NC	
PORT-13	NC	

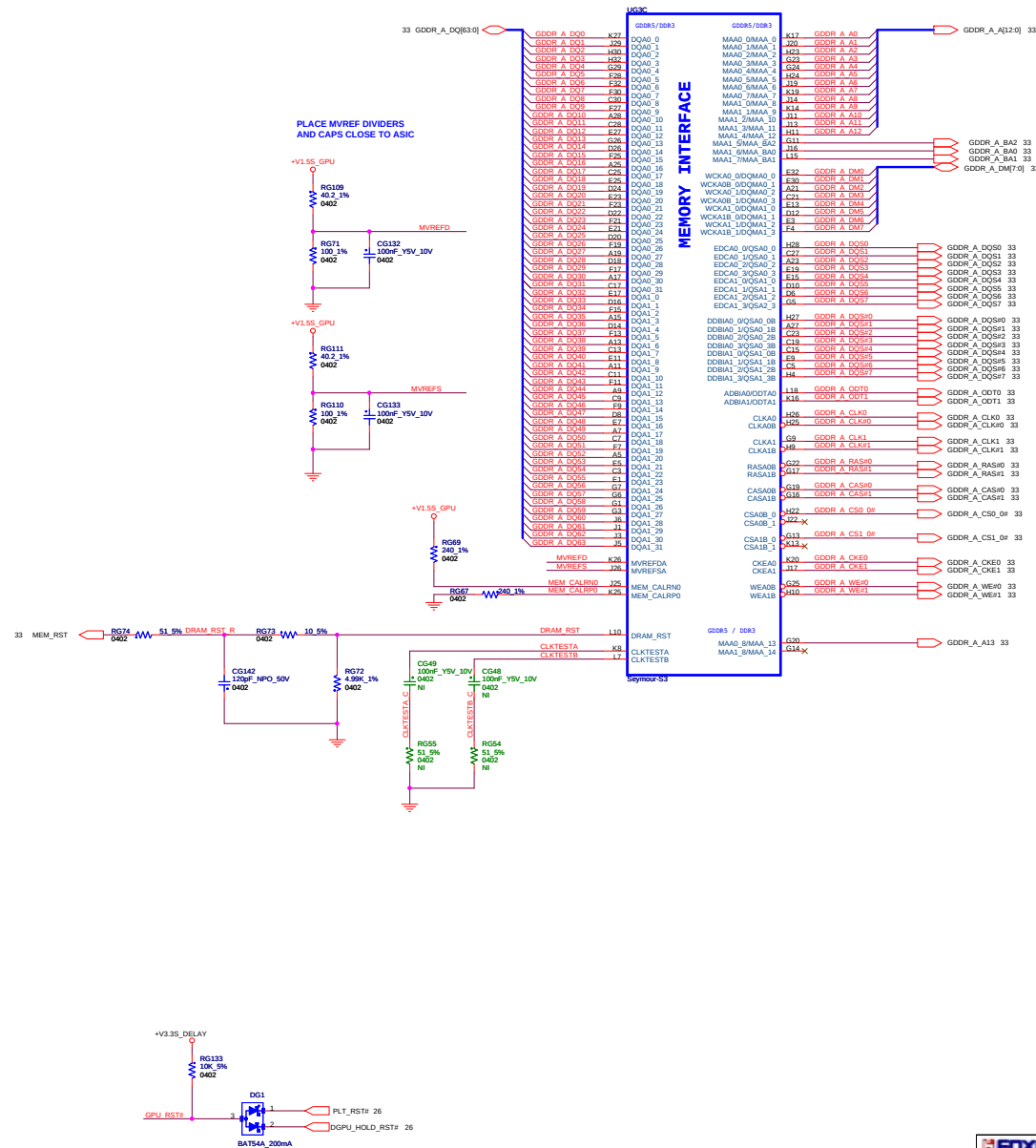
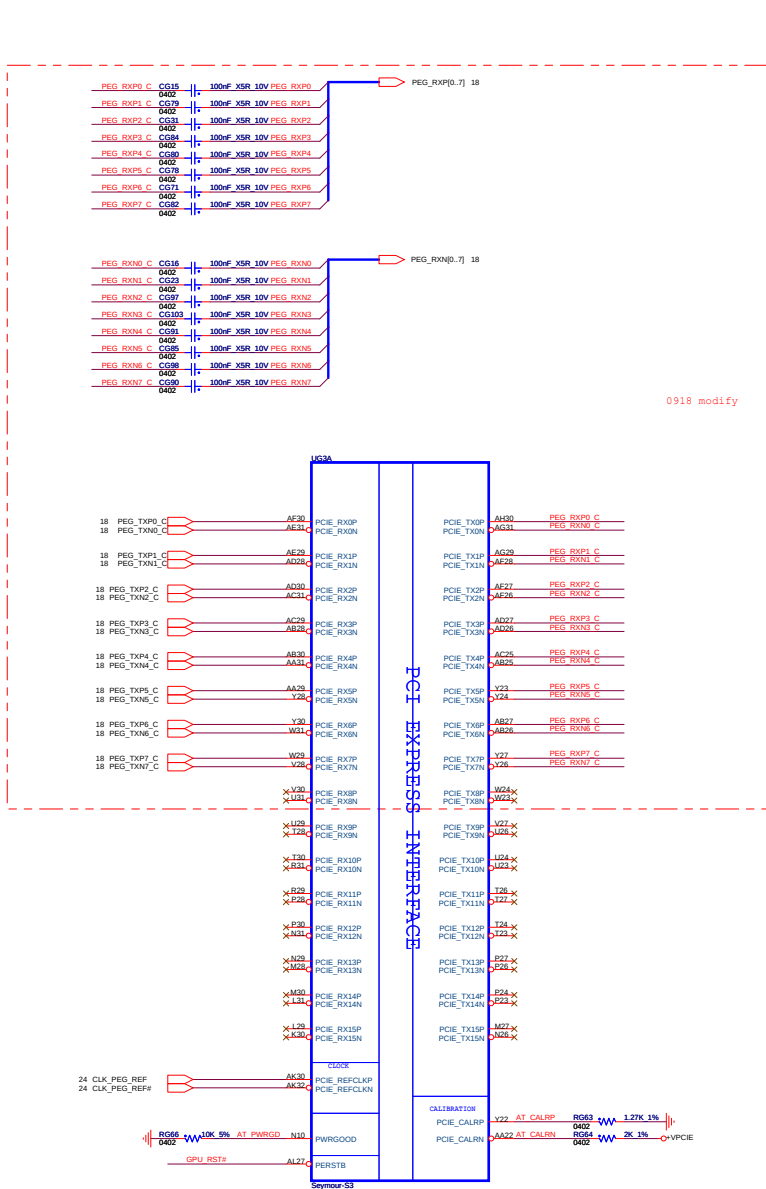


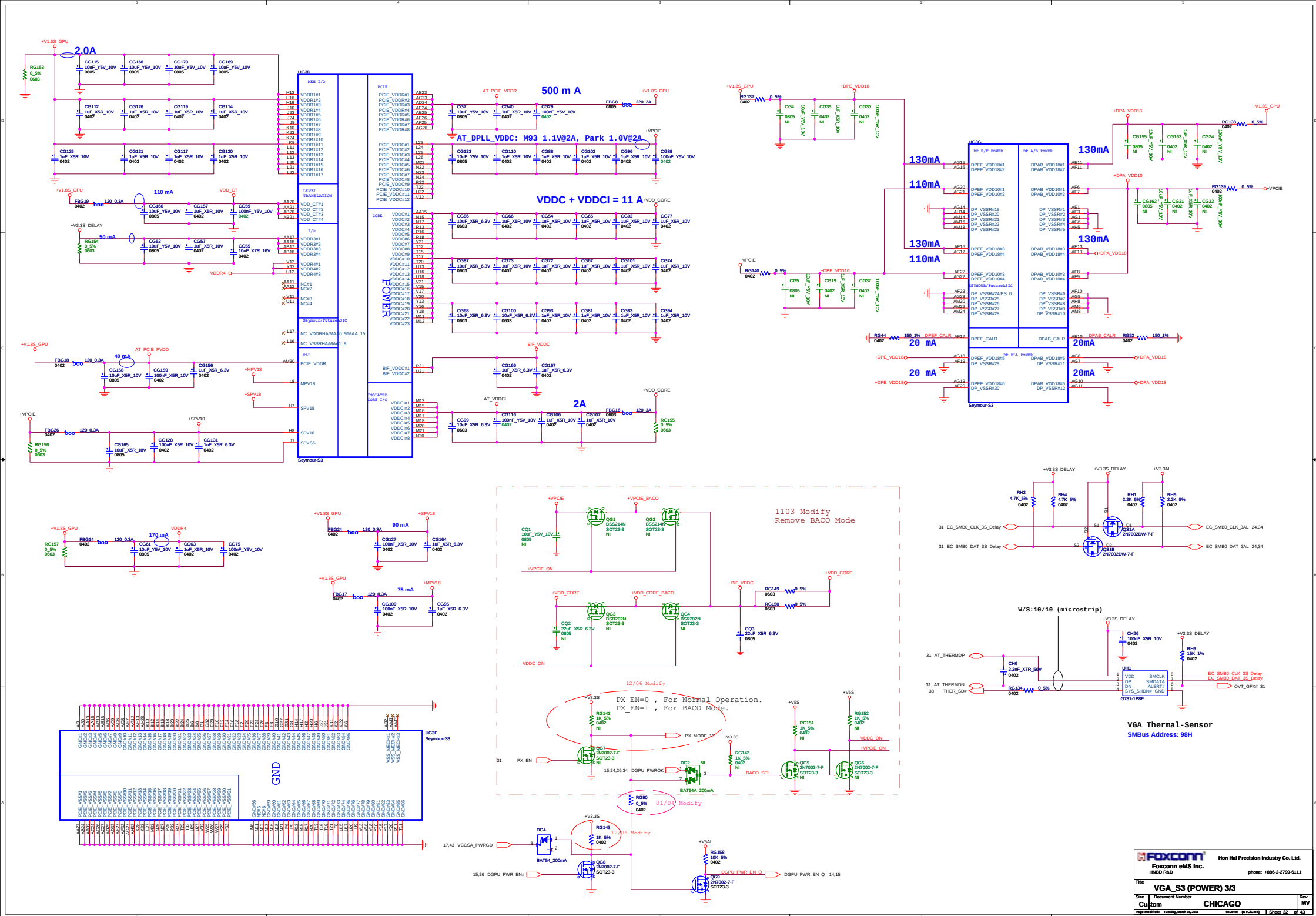
PLL ON DIE VR ENABLE	
GP28	0
	1
	disable
	Enable(Default)

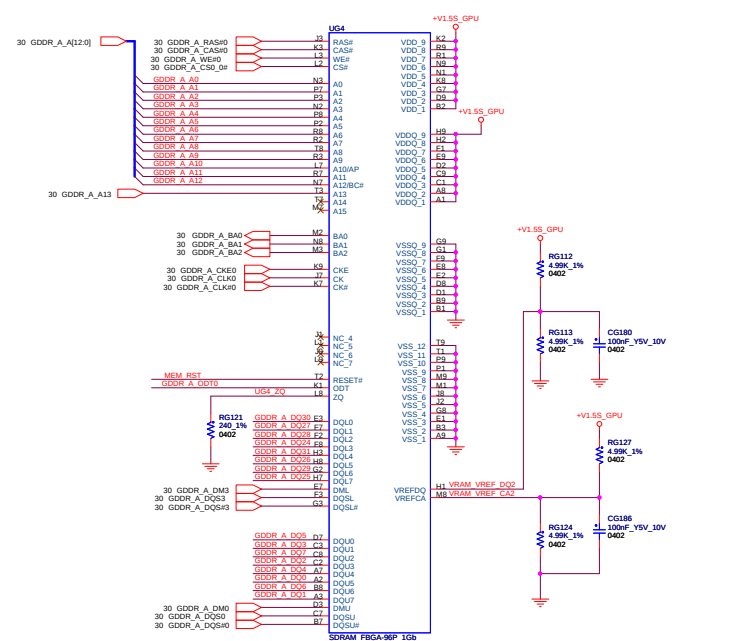
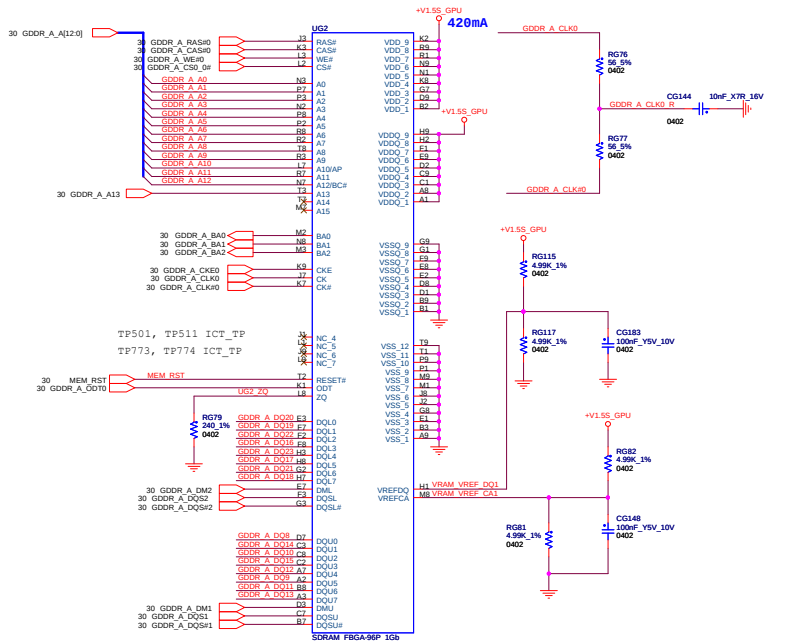




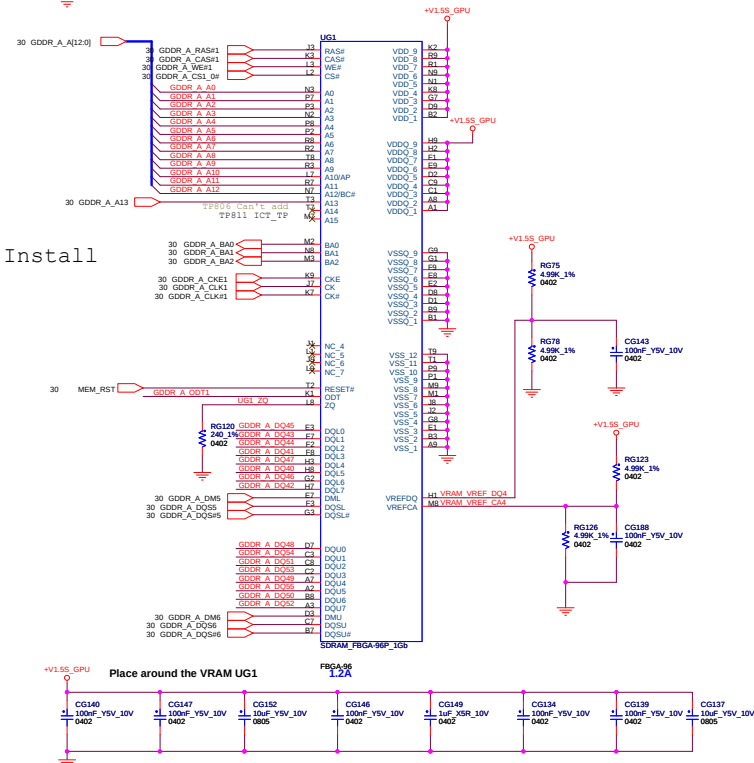
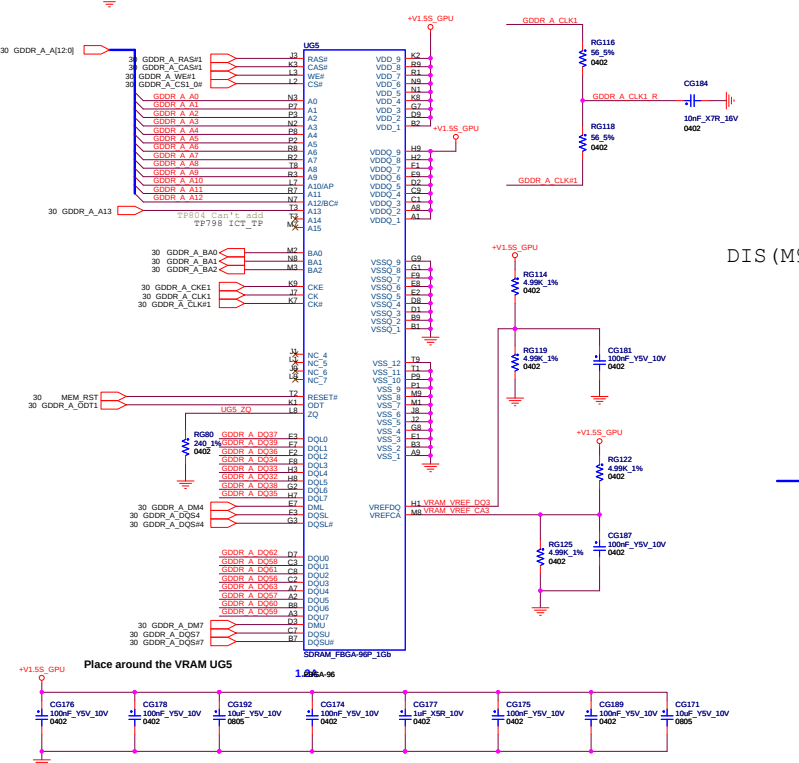


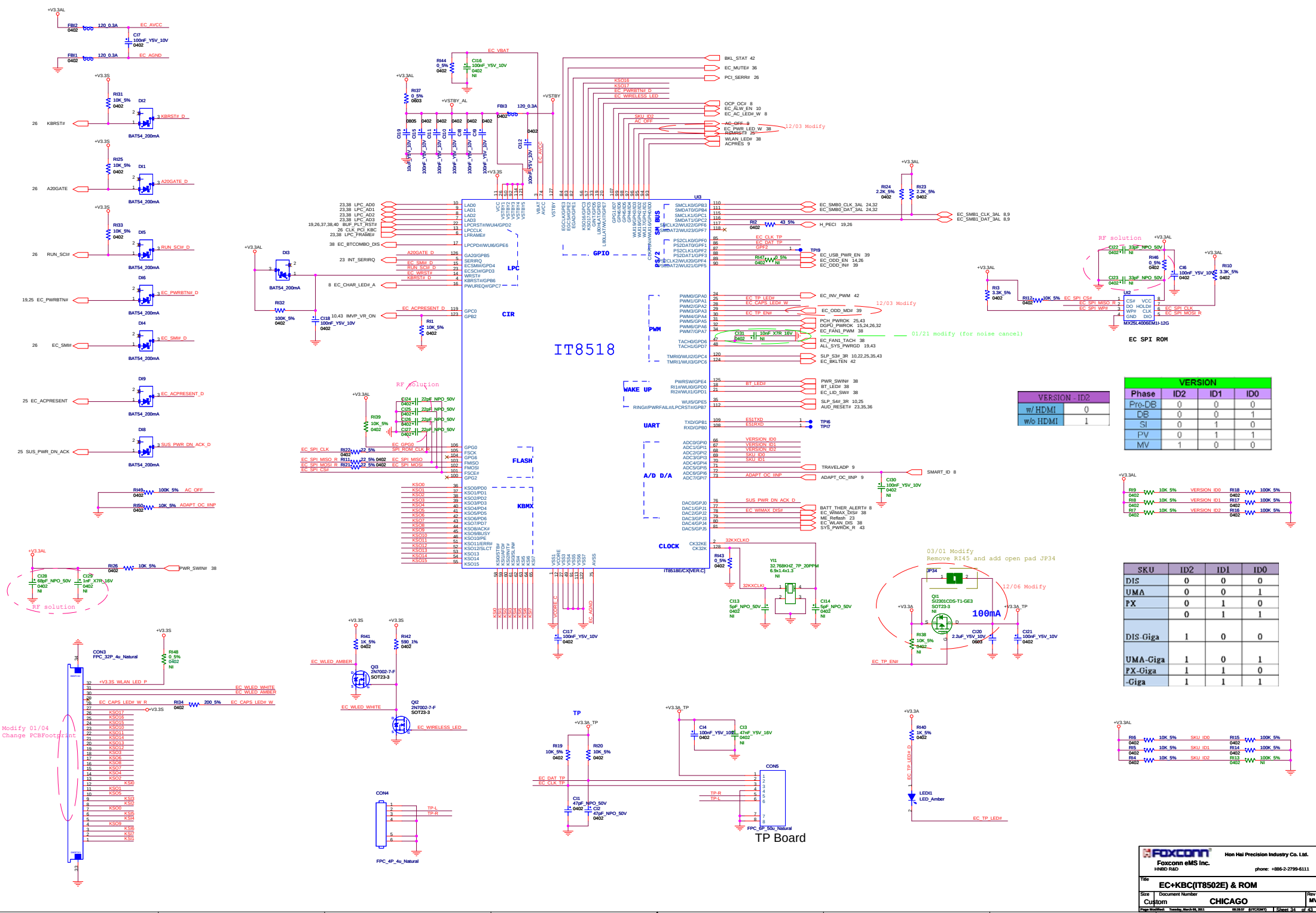






DIS(M93 XT);PX : Install



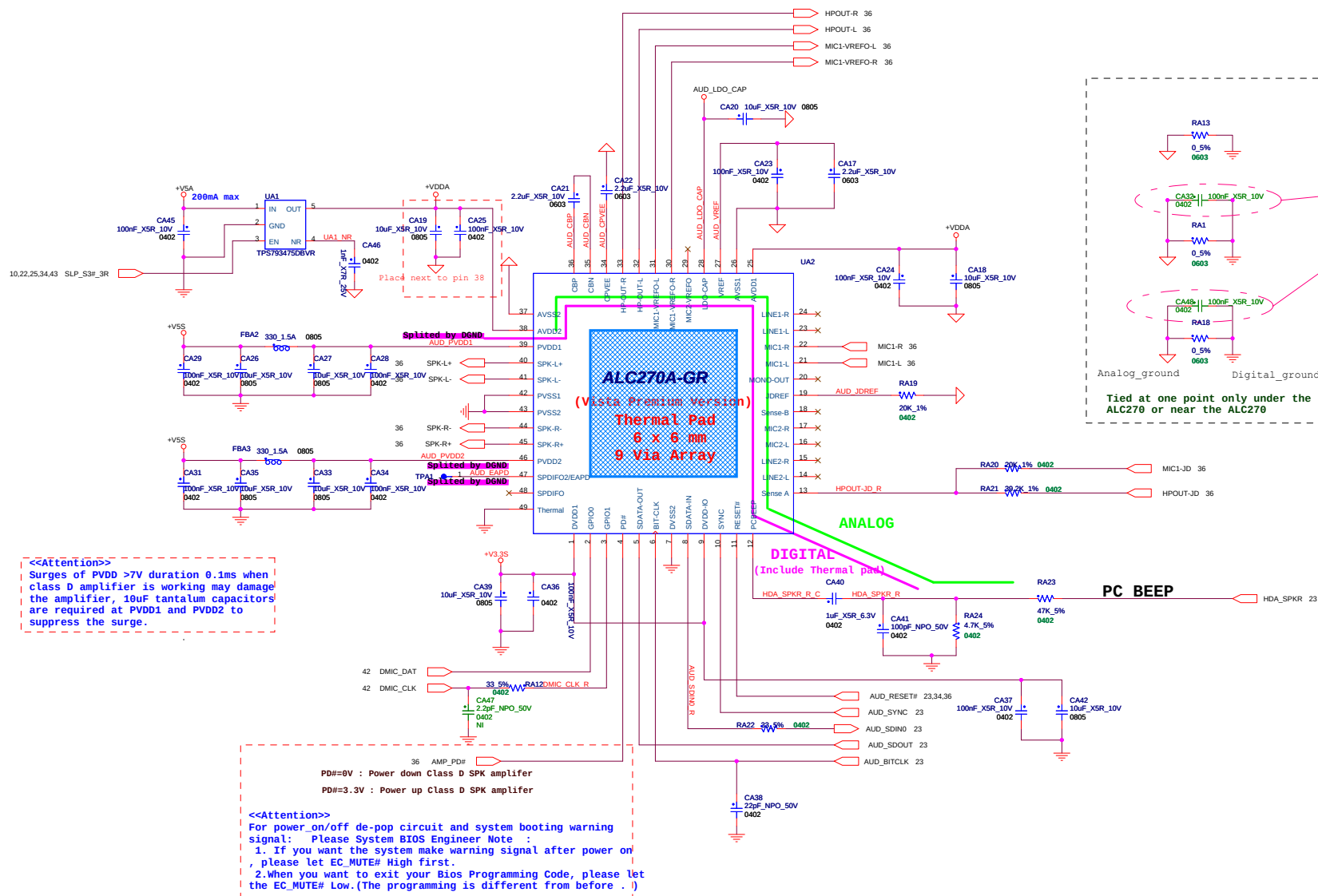


IT5518

VERSION ID2	
w/ HDMI	0
w/o HDMI	1

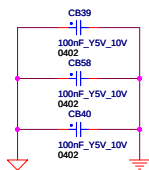
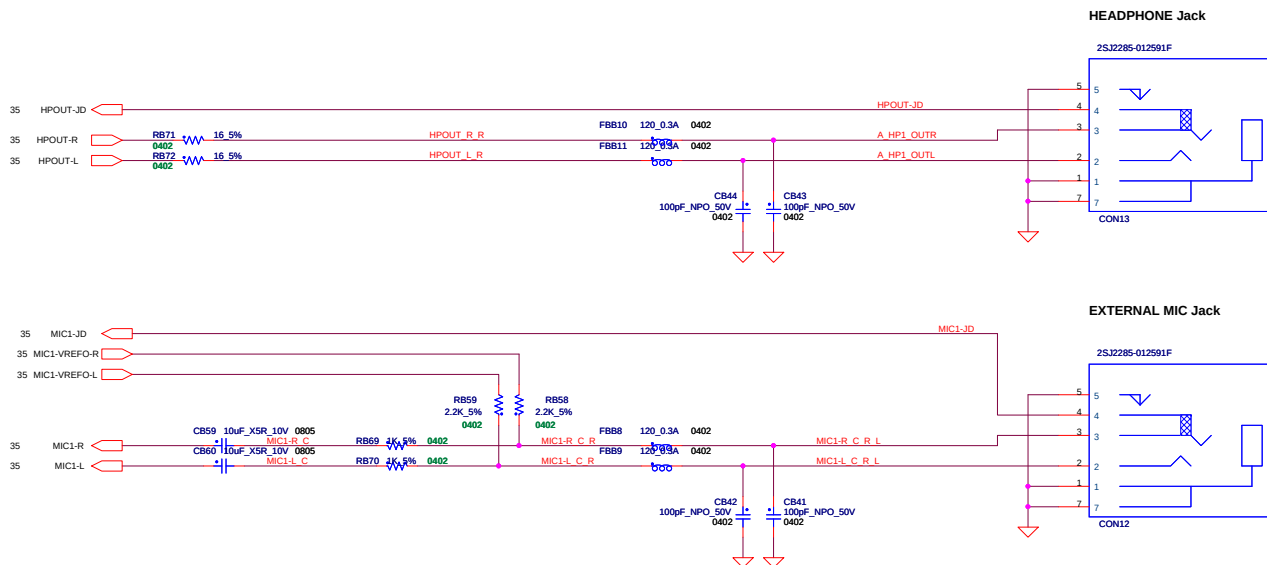
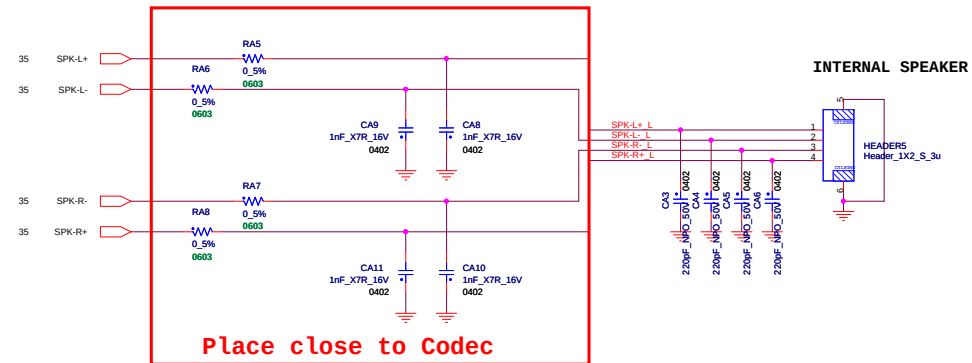
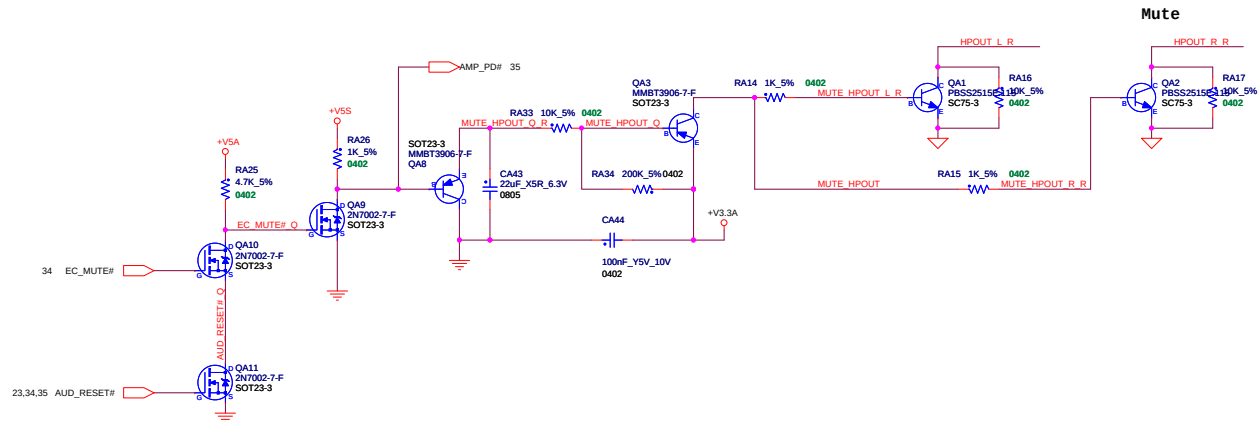
VERSION				
Phase	ID2	ID1	ID0	
Pre-DB	0	0	0	
DB	0	0	1	
SI	0	1	0	
PV	0	1	1	
MV	1	0	0	

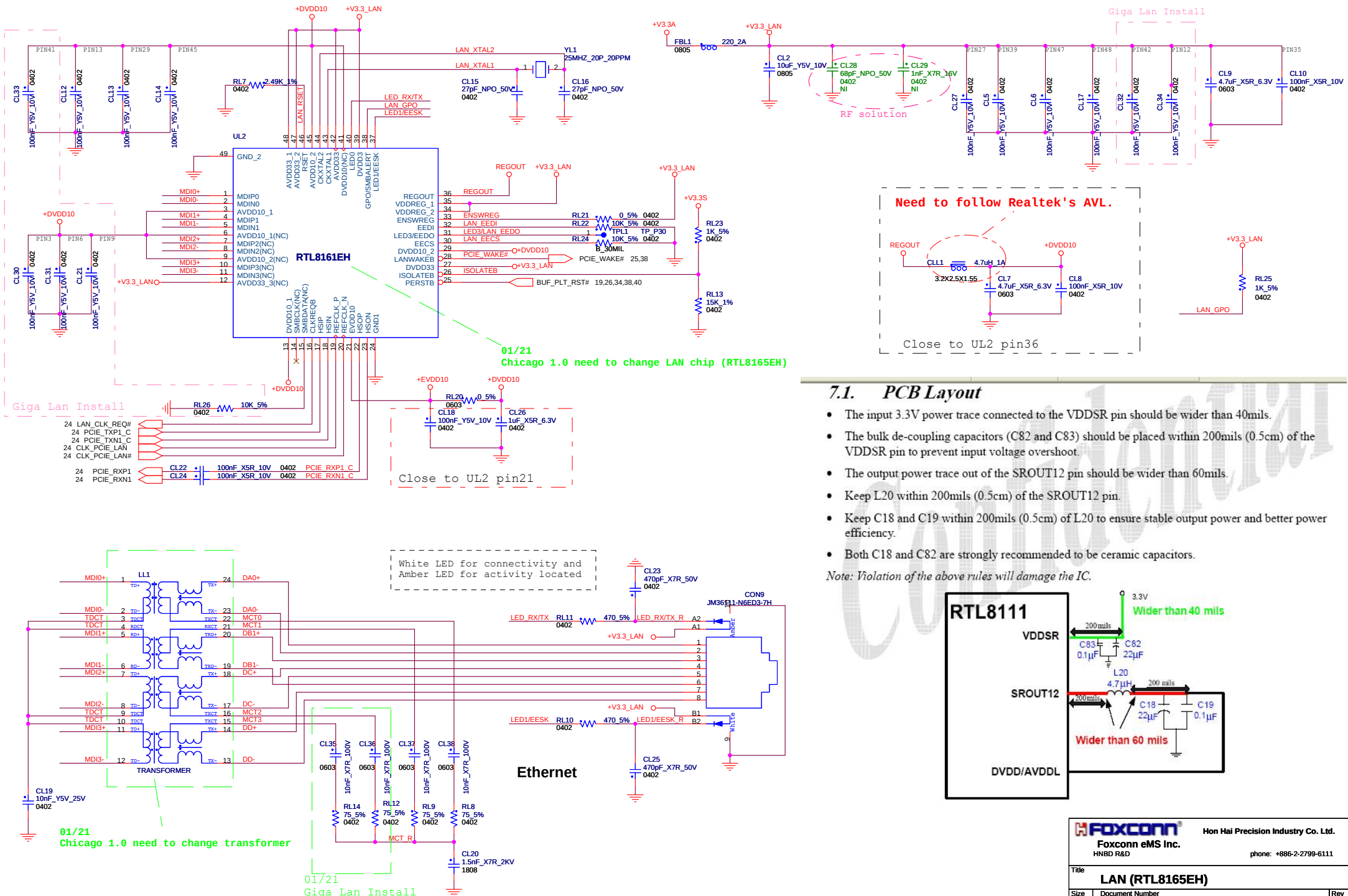
SKU	ID2	ID1	ID0
DIS	0	0	0
UMA	0	0	1
PX	0	1	0
	0	1	1
DIS Giga	1	0	0
UMA-Giga	1	0	1
PX-Giga	1	1	0
-Giga	1	1	1



<<Attention>>
Surges of PVDD >7V duration 0.1ms when
class D amplifier is working may damage
the amplifier. 10uF tantalum capacitors
are required at PVDD1 and PVDD2 to
suppress the surge.

<<Attention>>
For power_on/off de-pop circuit and system booting warning
signal: Please System BIOS Engineer Note :
1. If you want the system make warning signal after power on
, please let EC_MUTE# High first.
2. When you want to exit your Bios Programming Code, please let
the EC_MUTE# Low. (The programming is different from before .)

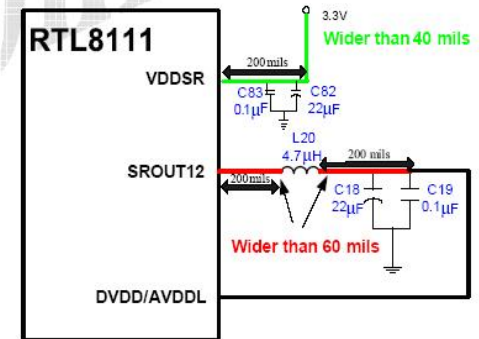


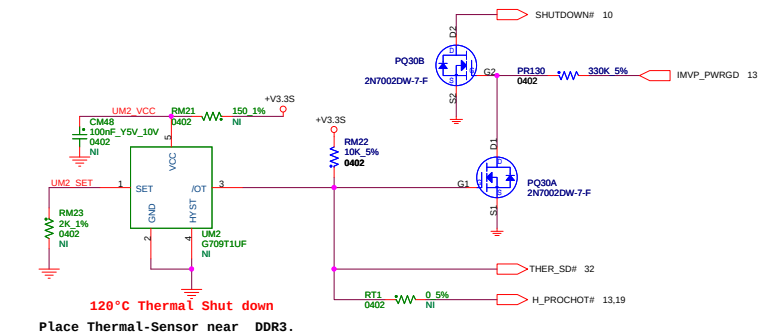


7.1. PCB Layout

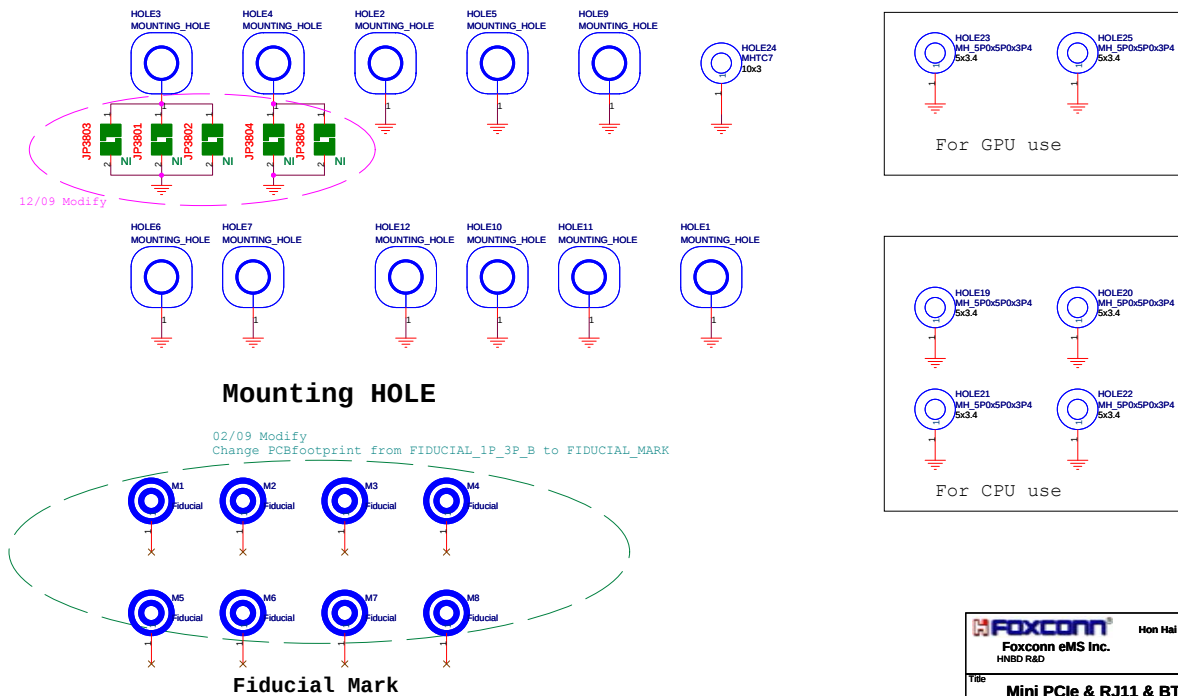
- The input 3.3V power trace connected to the VDDSR pin should be wider than 40mils.
- The bulk de-coupling capacitors (C82 and C83) should be placed within 200mils (0.5cm) of the VDDSR pin to prevent input voltage overshoot.
- The output power trace out of the SROUT12 pin should be wider than 60mils.
- Keep L20 within 200mils (0.5cm) of the SROUT12 pin.
- Keep C18 and C19 within 200mils (0.5cm) of L20 to ensure stable output power and better power efficiency.
- Both C18 and C82 are strongly recommended to be ceramic capacitors.

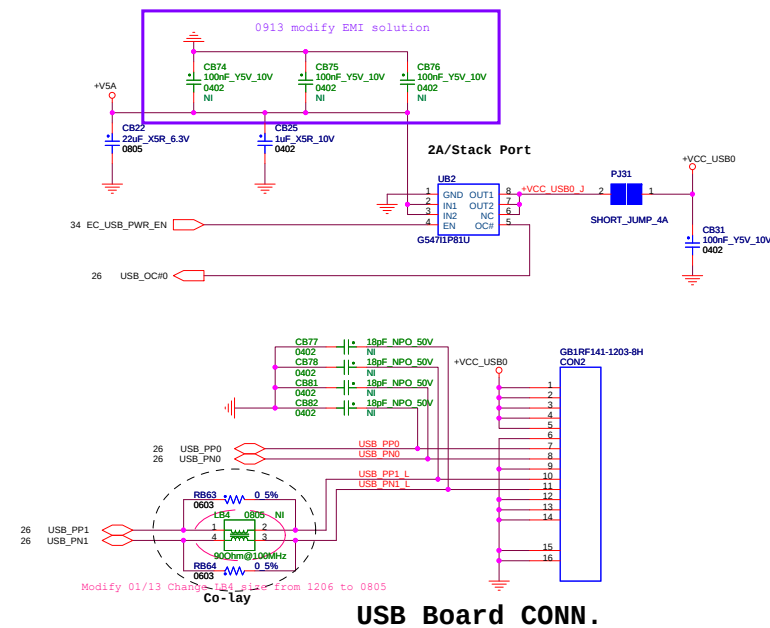
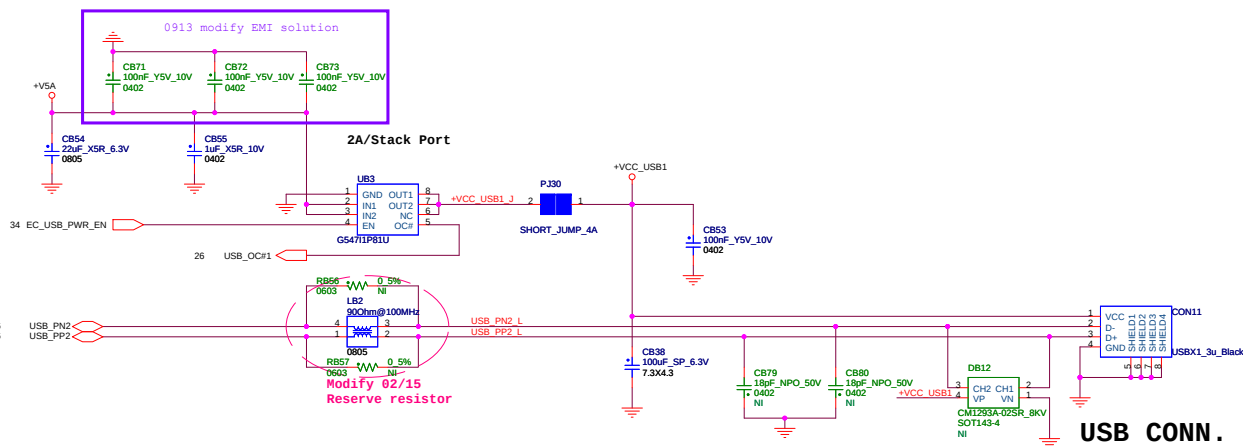
Note: Violation of the above rules will damage the IC.



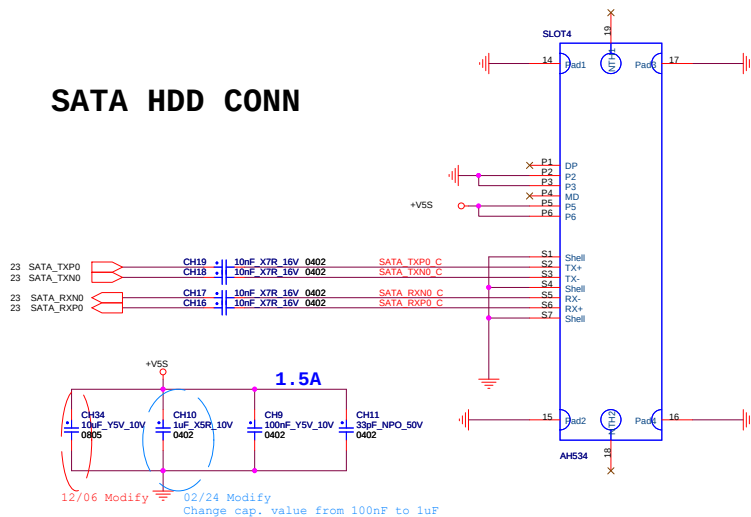


The schematic diagram shows the LED driver circuit for the EC_LID_SW16 pin. The circuit includes a 1202 LED, a 1202 LED driver IC, and various resistors and capacitors. The input is +V3.3AL FBH3. The output is EC_LID_SW16. The circuit is labeled "1202 Modify".

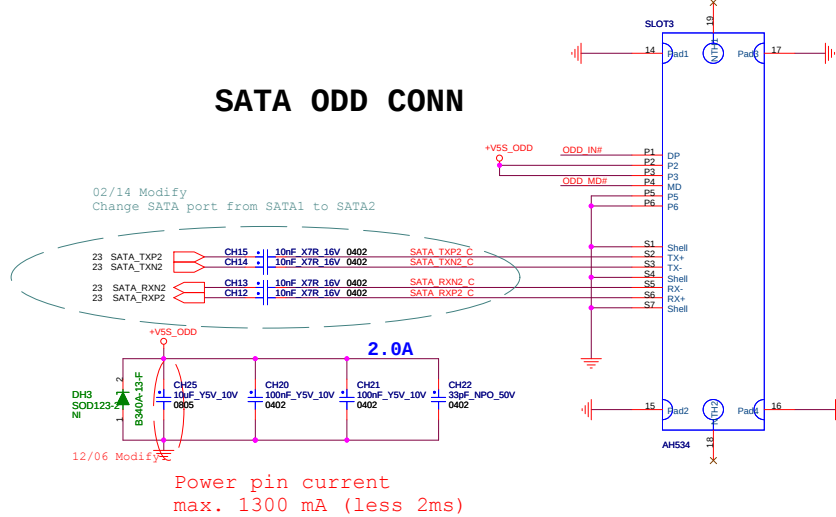




SATA HDD CONN



SATA ODD CONN



HDD/ODD Status LED

